
Pain

Dr. Muhannad Y. Al-Muhanna

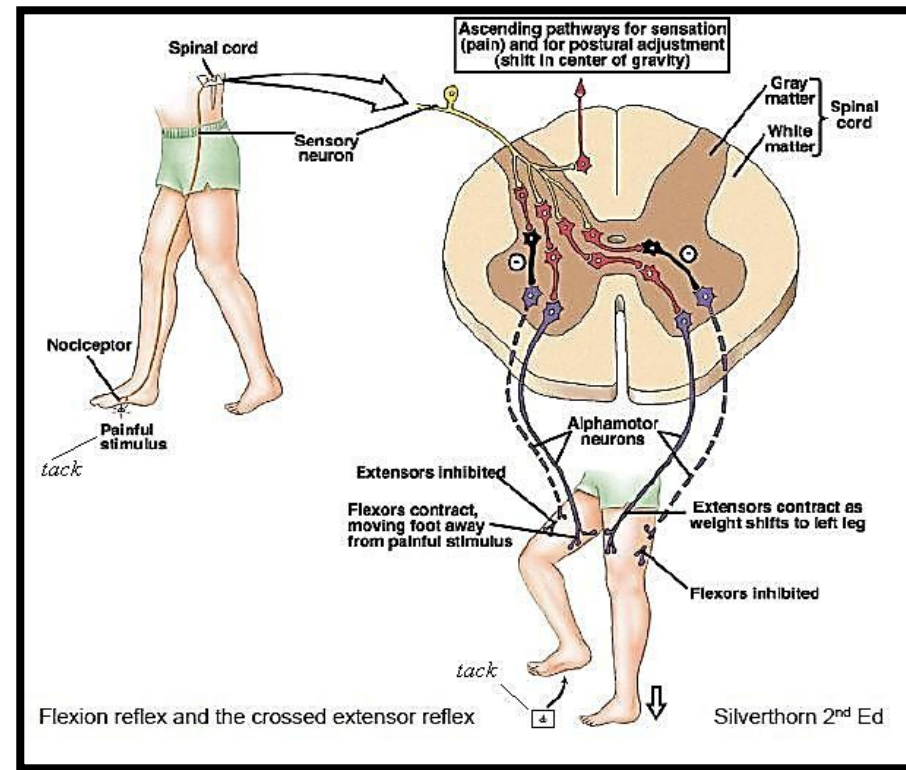
Definition of Pain

An unpleasant sensory and emotional experience associated with actual or potential tissue damage.

Is there any significance ?of pain

Protective Function

- It is a protective mechanism that make us aware that tissue damage is occurring or is about to occur:
 1. Avoid noxious stimuli.
 2. Remove body parts from danger.
 3. Promote healing by preventing further damage.
 4. Storage of painful experiences in memory helps us to avoid potentially harmful event in the future.



Significance of pain: why do we feel pain?

• Biological Function

The experience of pain may lead to the avoidance of potentially harmful situations and possible injury. Immobility and withdrawal

due to pain may serve to provide an environment in which healing and restoration

of function can occur.

-
- The sensation of pain may be accompanied by behavioral responses (withdrawal or defense) as well as emotional responses (crying or fear). (Note: negative emotions during the pain experience can enhance the strength and duration of memory for the pain).
 - Pain is perceived at both cortical and thalamic levels.
-

Threshold and Tolerance

- **Stimulus Threshold...** is the minimum point at which something (stimulus), such as pressure or heat, causes you **pain and is the same in each of us**
- **Tolerance ...** refers to the maximum amount of **pain** you can handle (or) **is our variable reaction to a painful stimulus.**

(it is affected by environment situation, psychological/emotional factors, increases with age, ongoing pain, placebo effect).

Pain receptors(nociceptors)

- ‘ Are special receptors that respond only to noxious stimuli and generate nerve impulses which the brain interprets as "pain".
- They are specific (have adequate stimulus) in that **pain is not produced by overstimulation of other receptors.**
- They are most widely distributed.
- Nociceptors are widespread in superficial layers of skin.
- Fewer in deep tissue and absent in brain tissue.

Pain receptors(nociceptors)

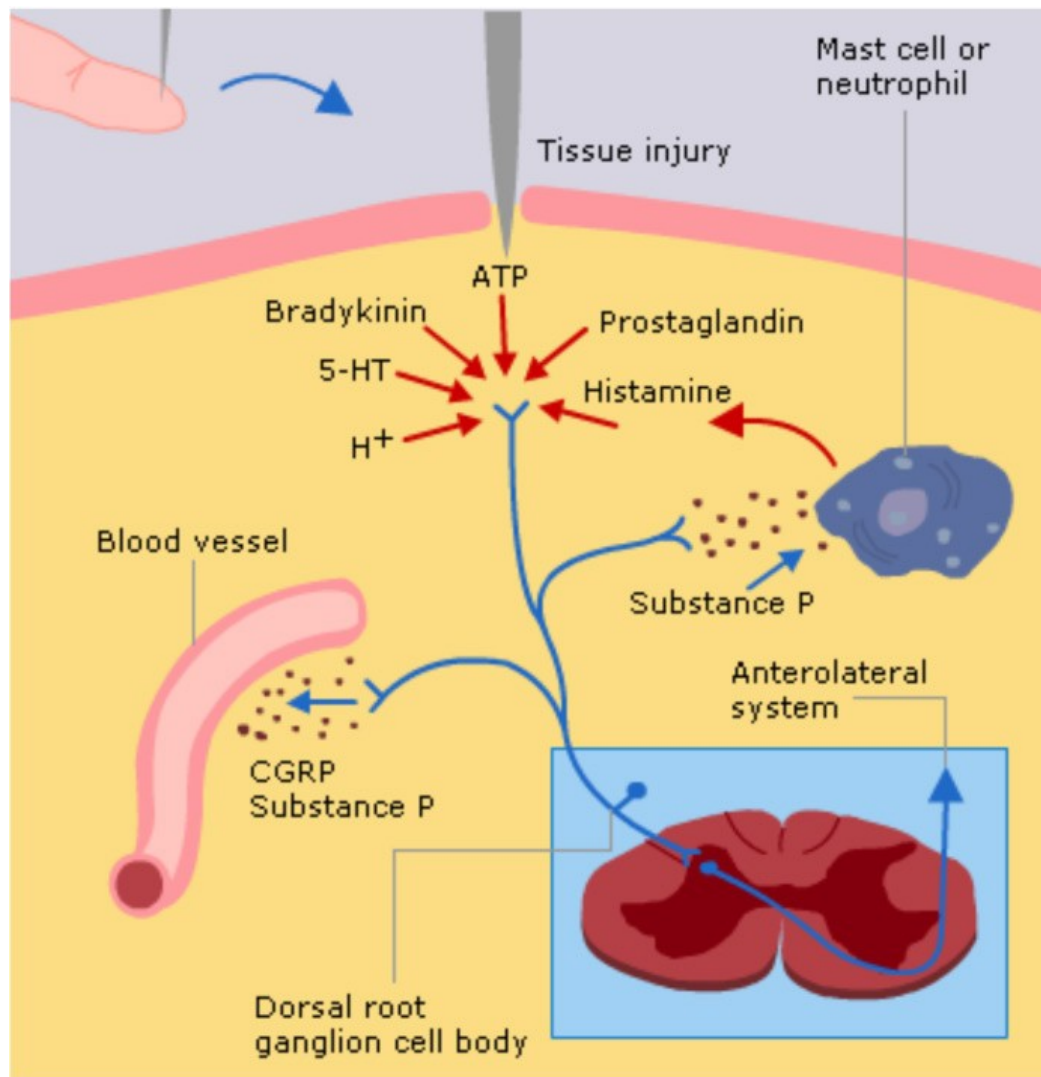
- They are found in largest number & density in skin, periostium joint surface, arterial wall & duramatar.
- pain receptors are activated by 3 types of stimuli:
 - 1. **Mechanical**- they elicit fast pain.
 - 2. **Thermal**- they elicit also fast pain.
 - 3. **Chemical**- they produce slow pain.

Pain receptors(nociceptors)

- They are high threshold receptors i.e. Painful stimuli must be strong and noxious to produce tissue damage.
- Heat pain threshold $> 43^{\circ}\text{C}$.
- Do not adapt (or very little) to repetitive stimulation (can be sensitized by various agents, (example: Prostaglandins)(it allows the pain to keep the person apprised of a tissue damaging stimulus as long as it persists).

Mechanism of stimulation of nociceptors

- Pain receptors are depolarized either directly or through the production of pain producing substances from damaged tissues as a result of inflammation (also called inflammatory mediators).
- Ex: Bradykinin, histamine, substance P, calcitonin gene - related peptide (CGRP), interleukins, prostaglandins (PGs), K^+ , Ach, proteolytic enzymes, acids.
- PGs & substance – P enhance the sensitivity of pain receptors.



Several types of nociceptor are described

- **Mechanoreceptors**, which respond to mechanical stimuli like pinch and pinprick.

- **polymodal nociceptors** are most prevalent and respond to excessive pressure, extremes of temperatures, and algogens (pain producing substances)

- **Thermal nociceptors**

- **Silent nociceptors**, which respond only in the presence of inflammation,

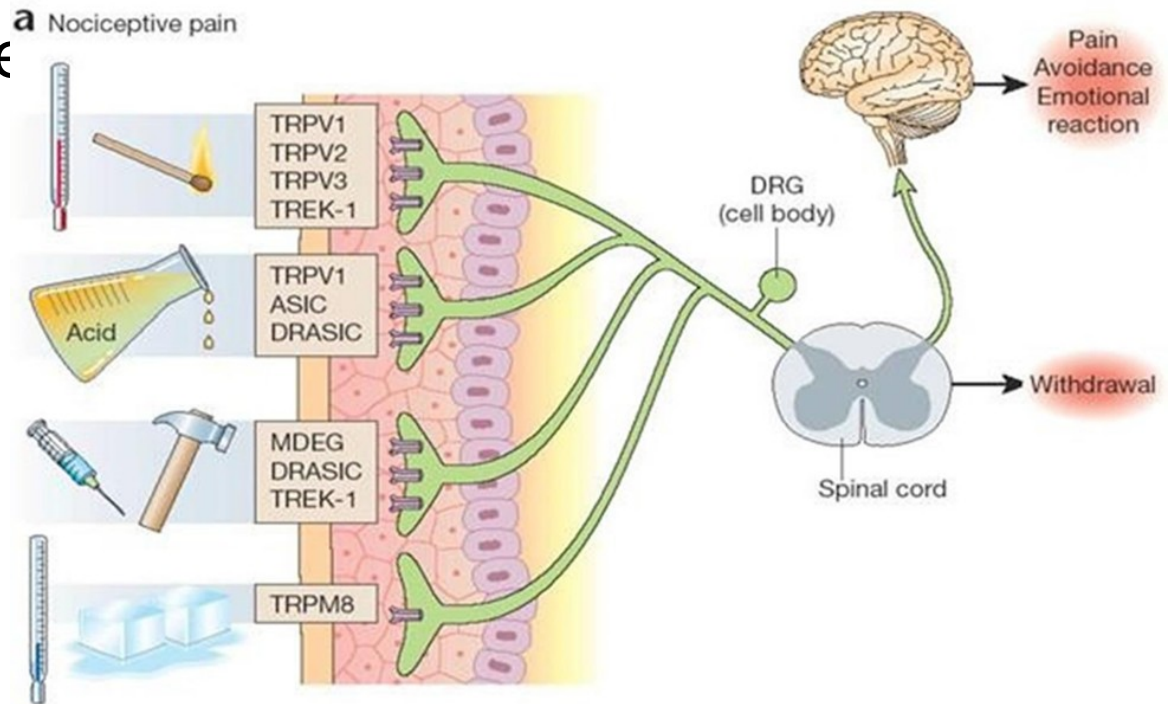
Polymodal nociceptors

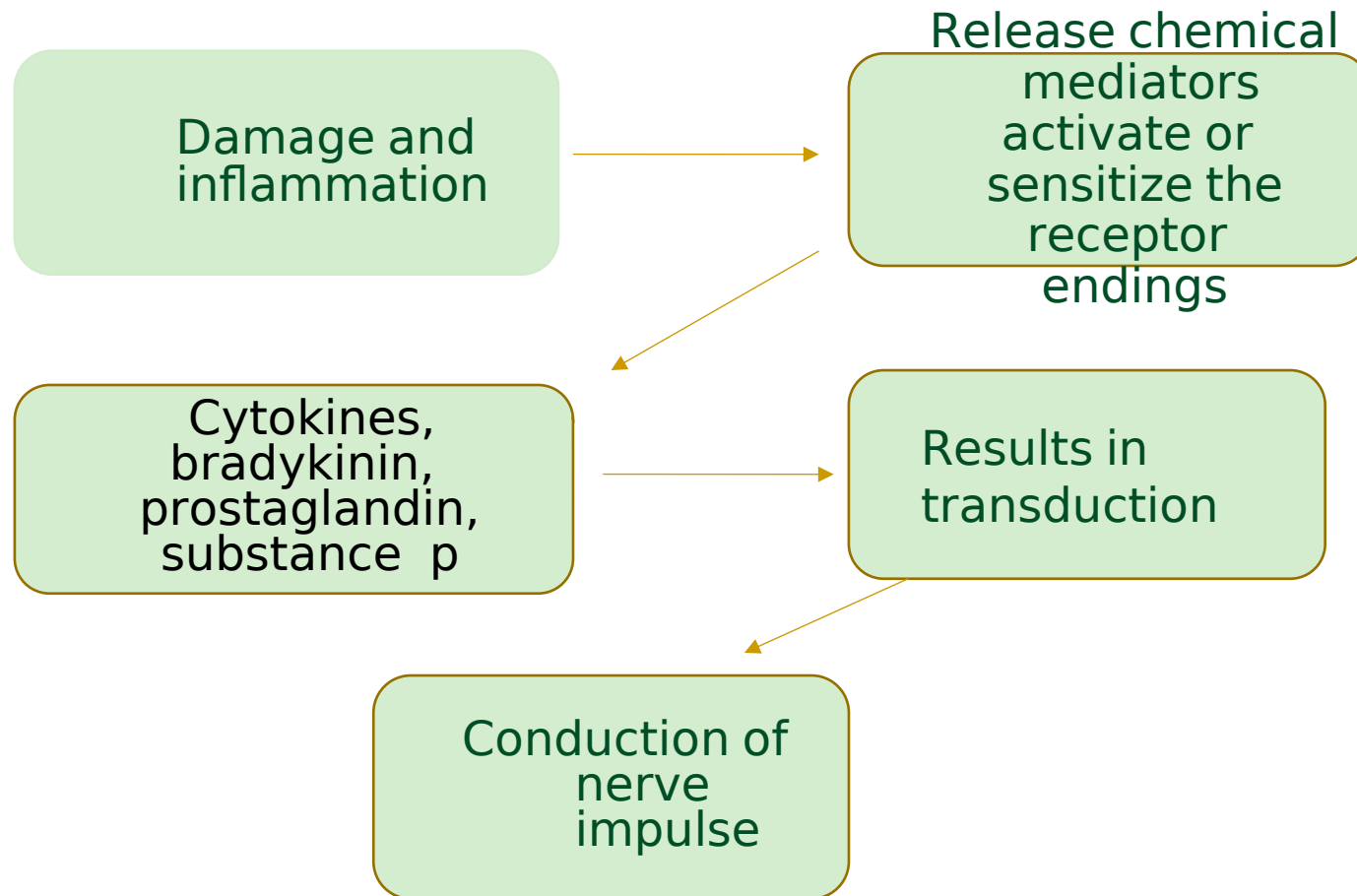
Polymodal nociceptors:

Mechanosensitive

Thermosensitive

Chemosensitive

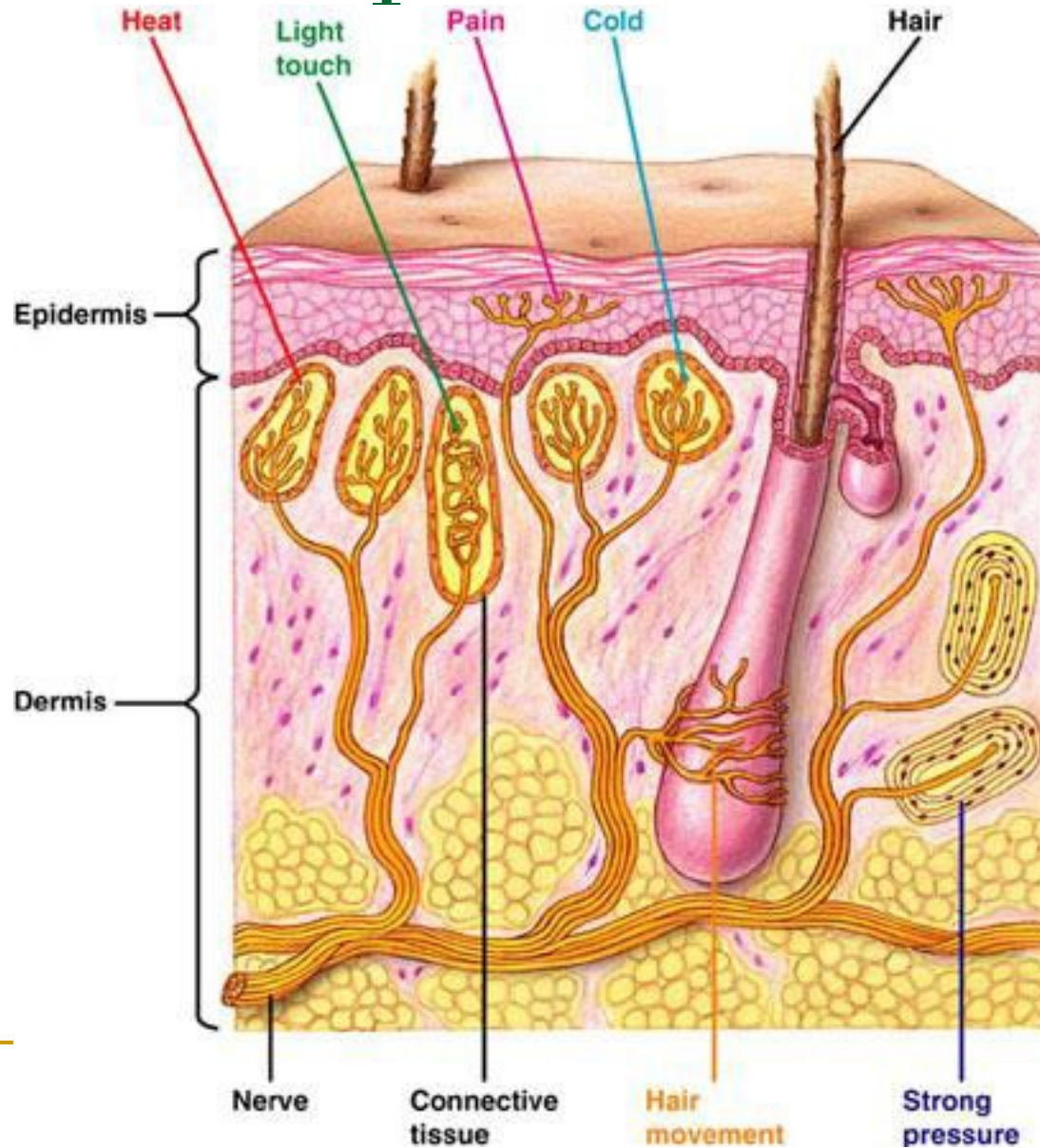




Chemical substance released during tissue damage

Substance	Source
K ⁺	Damaged cells
Serotonin	Platelets
Bradykinin	Plasma
Histamine	Mast cells
Prostaglandins	Damaged cells
Leukotrienes	Damaged cells
Substance P	Primary nerve afferents

:Pain Receptor



إِنَّ الَّذِينَ كَفَرُوا بِآيَاتِنَا سَوْفَ نُصْلِيهِمْ نَارًا كُلَّمَا
تَصَوَّغُوا فِيهَا تُبَدِّلْنَاهُمْ جُلُودًا غَيْرَهَا لِيَذُوقُوا
((الْعَذَابَ قَلِيلًا إِنَّ اللَّهَ كَانَ عَزِيزًا حَكِيمًا))
سورة النساء (56)

Third-degree burn, or full-thickness, burns destroy the entire thickness of the skin. **Mostly there is no pain in the area of the burn**, because the pain receptors have been obliterated along with the rest of the dermis.

:Pain (Nociceptive) Processing

- *The physiologic process of pain sensation is termed nociception.*

- **Nociception:** *Non-conscious neural transmission of signals evoked by activation of nociceptors (or originating with tissues trauma) from periphery to the CNS.*

- *it consists of the following processes :*

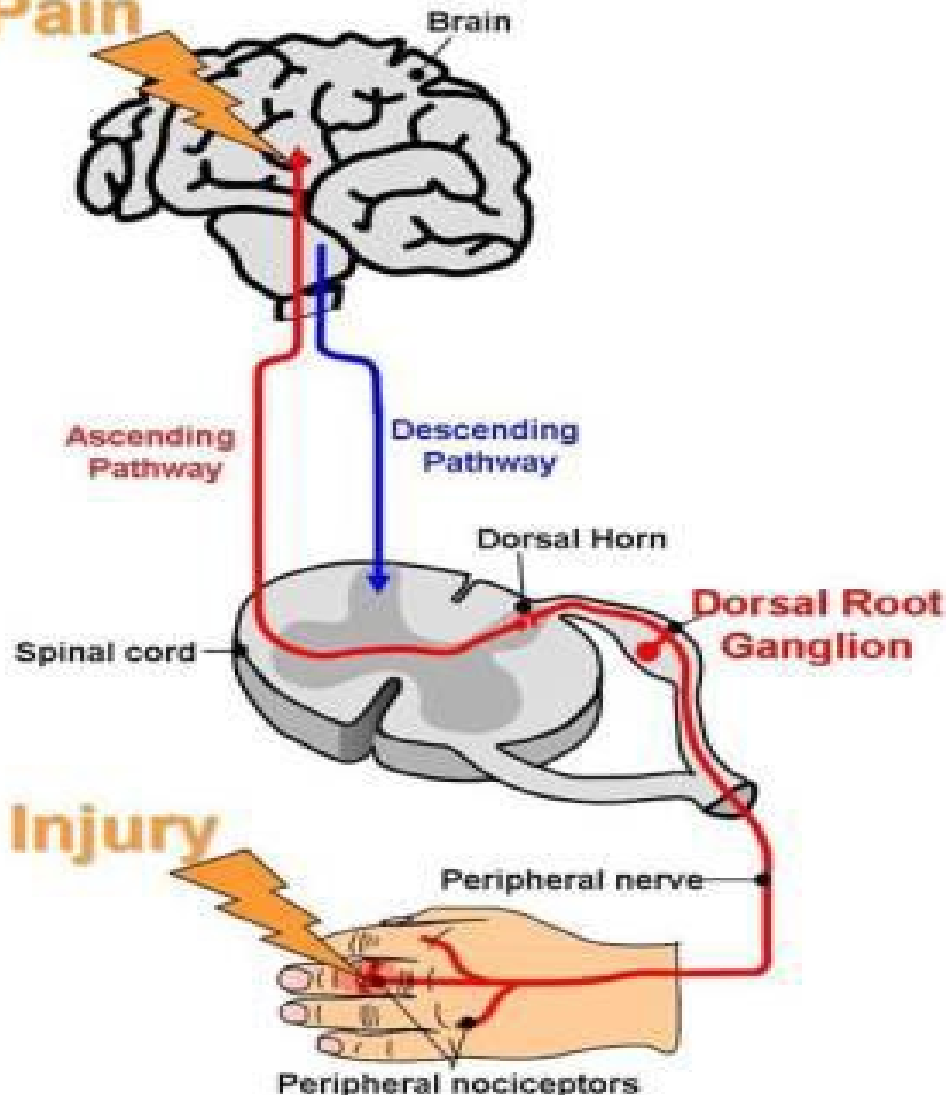
- *Transduction (at the level of receptors)*

- *Transmission*

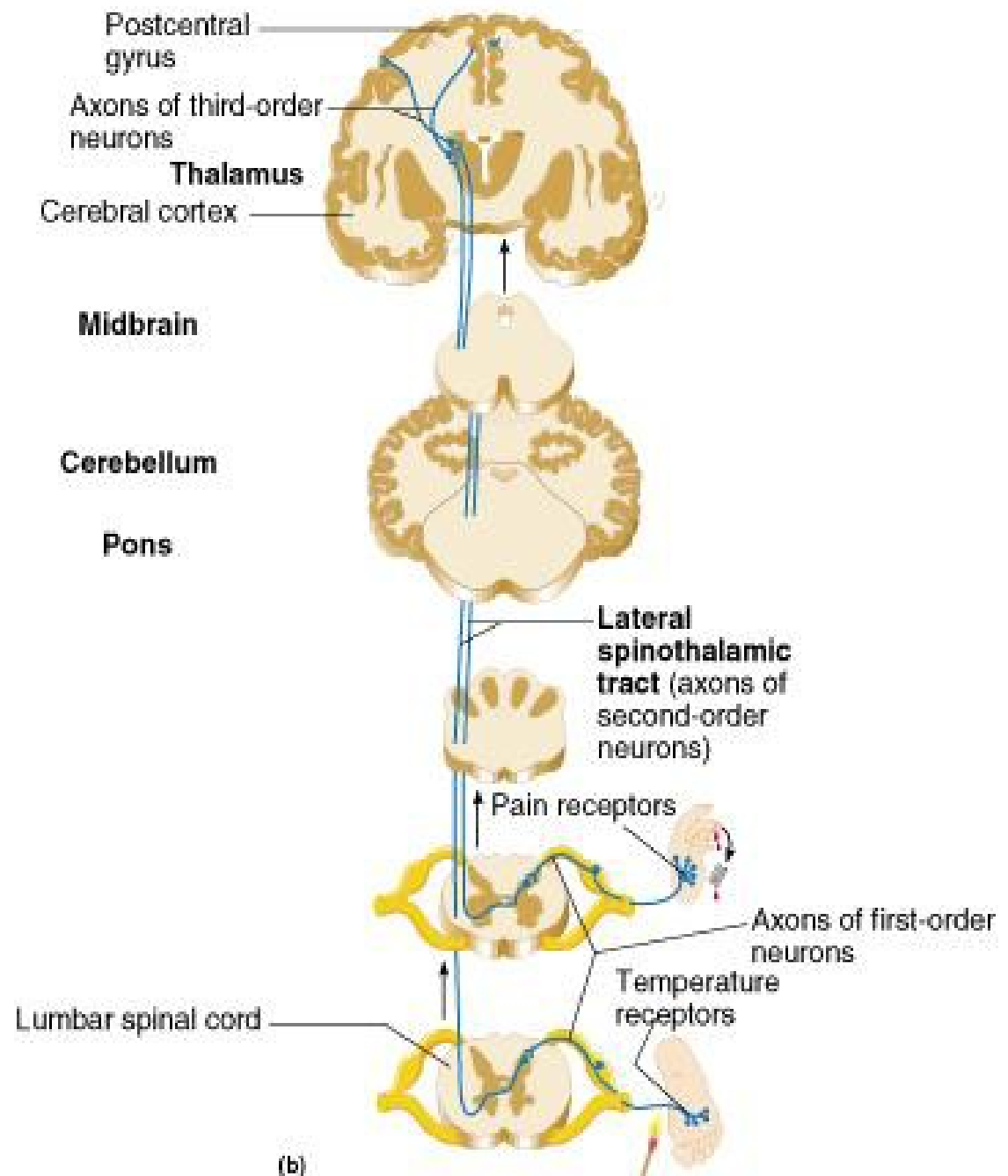
- *Modulation*

- *Perception*

Pain



Pain Pathways: Review



Stages of Nociception

- transduction:

activation of nociceptors by a stimulus (convert stimulus to nerve impulses).

- transmission:

relay of action potentials along nociceptive fibres to CNS

- modulation:

by either peripheral nerves or CNS mechanisms

- perception (where pain is felt)

the interpretation by the brain of the sensation as pain.

:Pain Transduction

- *Nociceptors: Latin – noci = harm or injury)*
- *is used only to describe the neural response to traumatic or noxious stimuli.*
- *Most nociceptors are free nerve endings that sense heat, mechanical and chemical tissue damage.*

Types of nerve fibers:

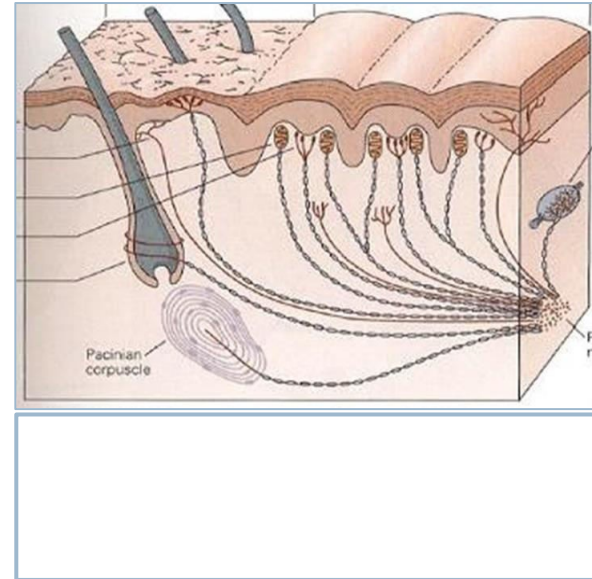
Myelinated (a-fiber).

A α (thickly myelinated).

A β (intermediate m.).

A δ (thinly myelinated).

Unmyelinated (c-fiber).



All pain receptors are free nerve endings of unmyelinated type C fibers and small diameter type A δ fibers.

Transmission of pain

•Pain impulses are transmitted by two fiber systems. ($A\delta$ and C fibers),

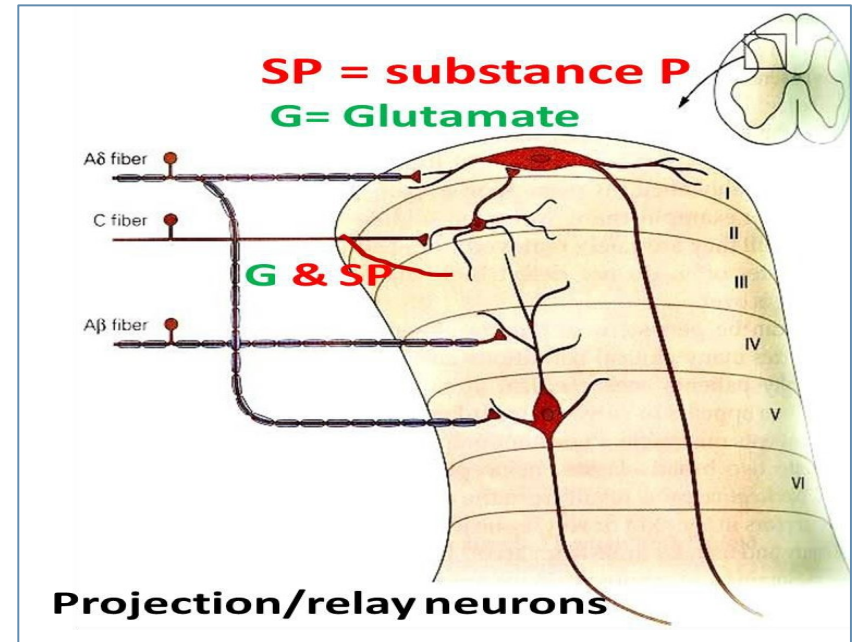
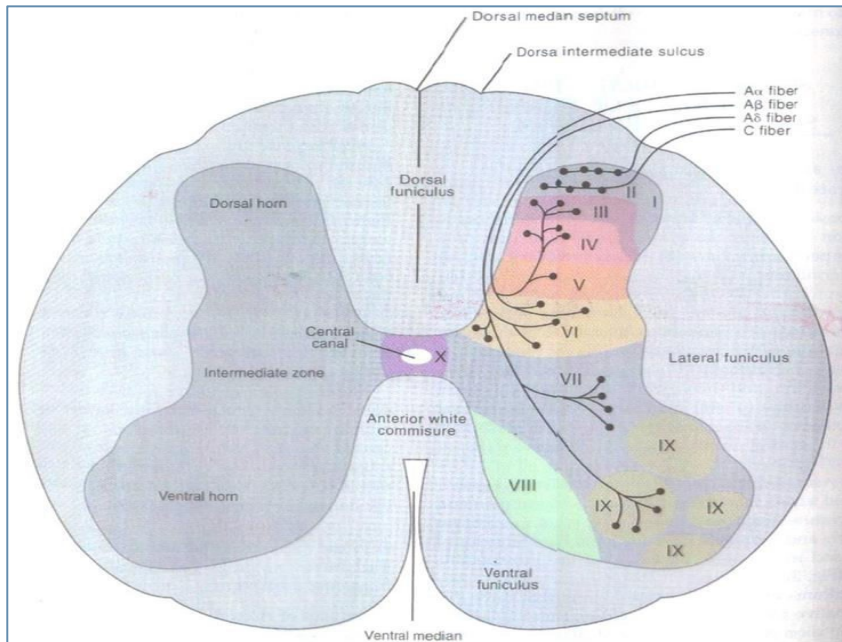
C	$A\delta$
<i>Characteristic</i> •Primary afferent fiber •Small diameter •Unmyelinated •Slow conducted <i>Pain quality:</i> •Diffuse, dull, slow, poorly localized, chronic pain	<i>Characteristic</i> •Primary afferent fiber •Larger diameter •myelinated •fast conducted <i>Pain quality:</i> •Well localized, sharp, fast, well localized, acute pain

Pain Transmission (spinal network)

- ***A δ fibers** terminate in the most superficial layer in spinal cord, lamina I (also called the marginal zone), with some fibers projecting more deeply to lamina V.*
- ***C fibers** are also destined for the superficial dorsal horn, with the focus in lamina II and III (the substantia gelatinosa)*

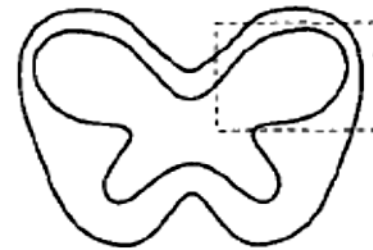
Projection neurons: are Dorsal Horn Neurons projecting pain encoded information to contralateral thalamic nuclei.

Nociceptive input to the spinal cord

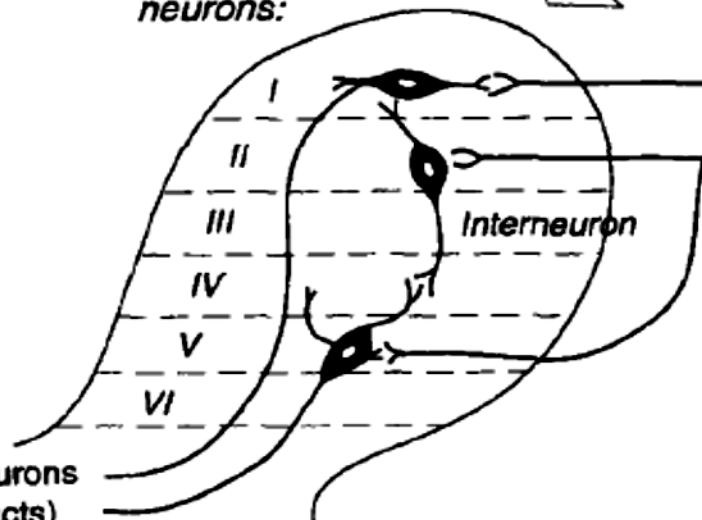


- Aδ-fibers terminate in **lamina I and V (fast pain)** and neurotransmitter is **glutamate**.
- C-fibers terminate in **lamina II and III (substantia gelatinosa)** (slow pain) and neurotransmitter is **substance P**

*Spinal cord:
(in cross section)*



*Dorsal horn
neurons:*



*Primary
afferents:*

A delta fiber

C fiber

Pain pathway (Ascending spinal tract)

- **Spinothalamic tract:** *(direct pathway of the anterolateral system) for acute, localized pain*

- **Spinoreticular tract:** *(indirect pathway of the anterolateral system) for slow chronic, dull pain*

Dual Pain Pathways

Pain signals take 2 pathways to the brain.

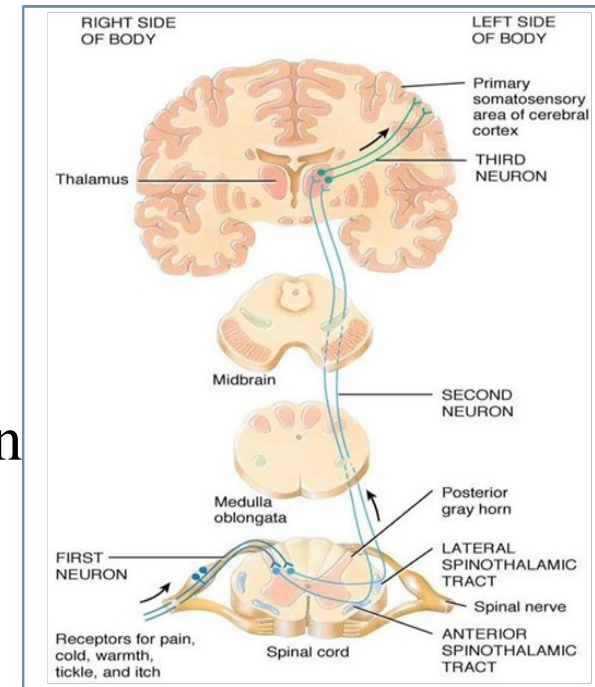
Most of the slow pain fibers project to reticular formation & then proceed to thalamus (posterior nuclei) . Reticular system project to all parts of brain but specially to cerebral cortex therefore they cause arousal from sleep.

1. Neospinothalamic (lateral STT):

- Fast (acute) pain ($\alpha\delta$ -type).
- Thermal pain (acute type).

2. Paleospinothalamic (ventral STT):

- Slow pain (c-type) plus some $\alpha\delta$. Chronic pain
- Crude touch & pressure.
- Itch & tickle.
- Sexual sensations.



Where do fibers of lateral & ventral STT ?terminate

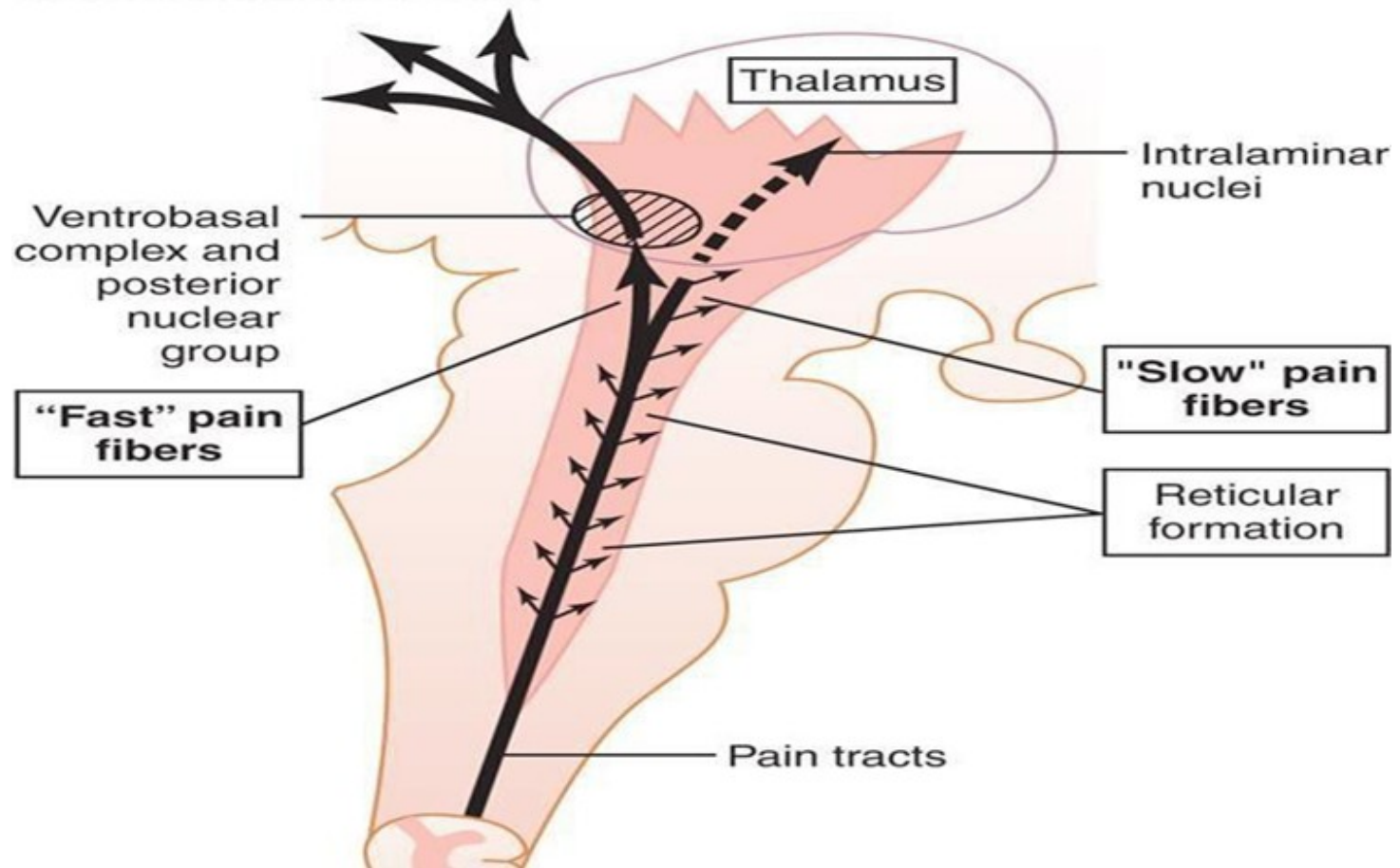
Most pain fibers of **lateral STT (L-STT)** pass all the way to thalamus:

- Ventrobasal complex.
- Posterior nuclear group.

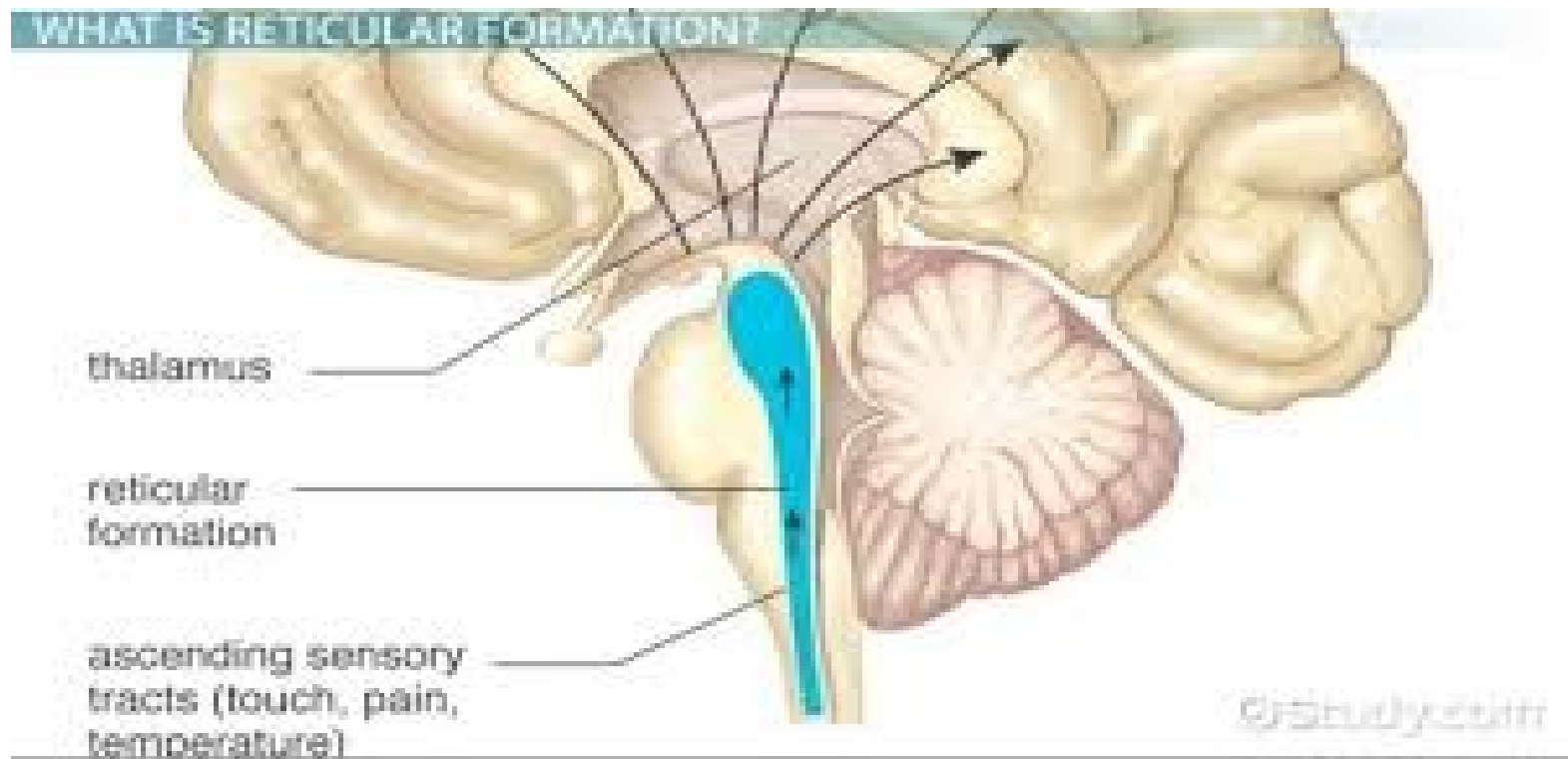
Only few pain fibers of **ventral STT** pass all the way to thalamus:

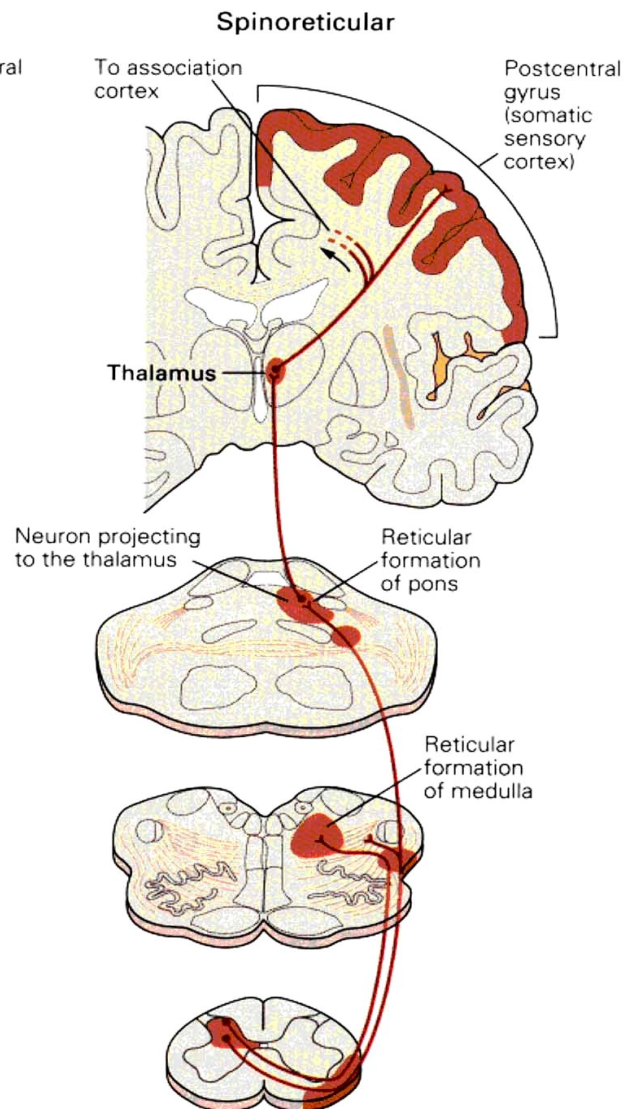
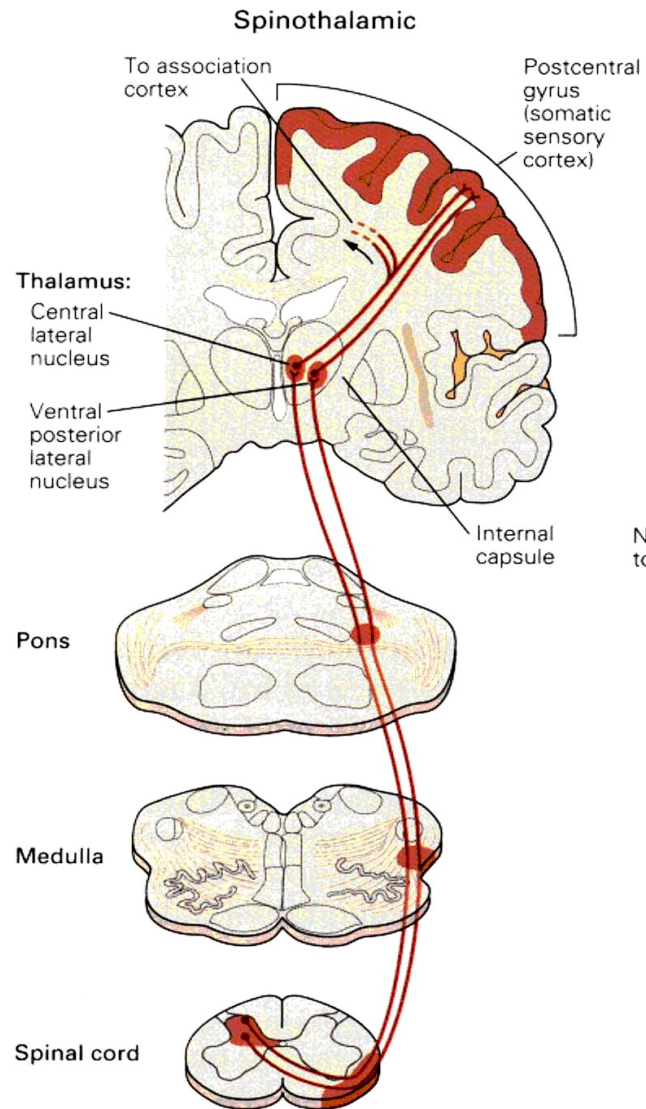
- they project to Brain stem reticular formation.
-

To: Somatosensory areas

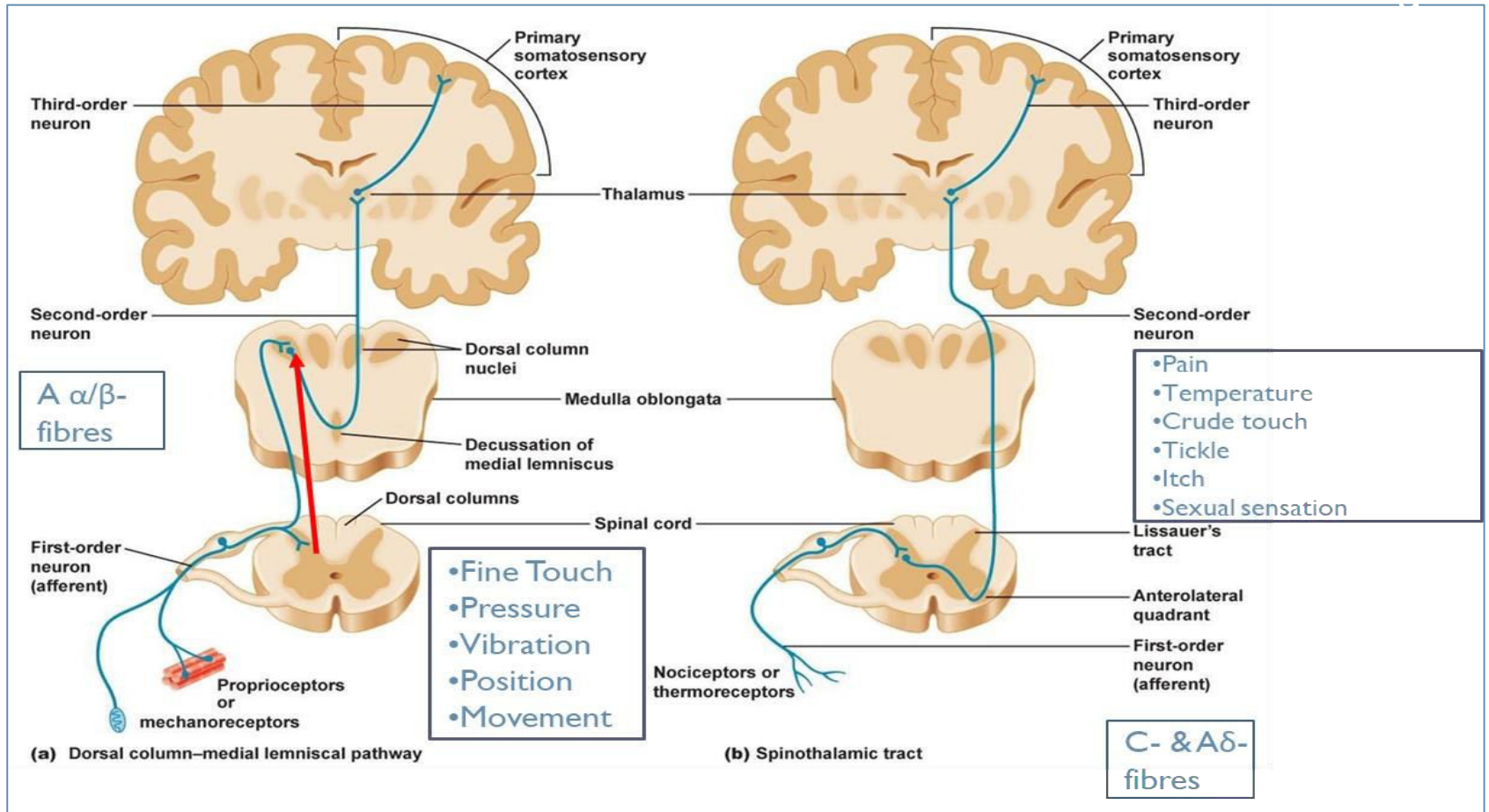


The reticular formation is a complex, net-like network of neurons in the brainstem that acts as a crucial relay and integration center, regulating vital functions like wakefulness, attention, sleep-wake cycles, and autonomic control.

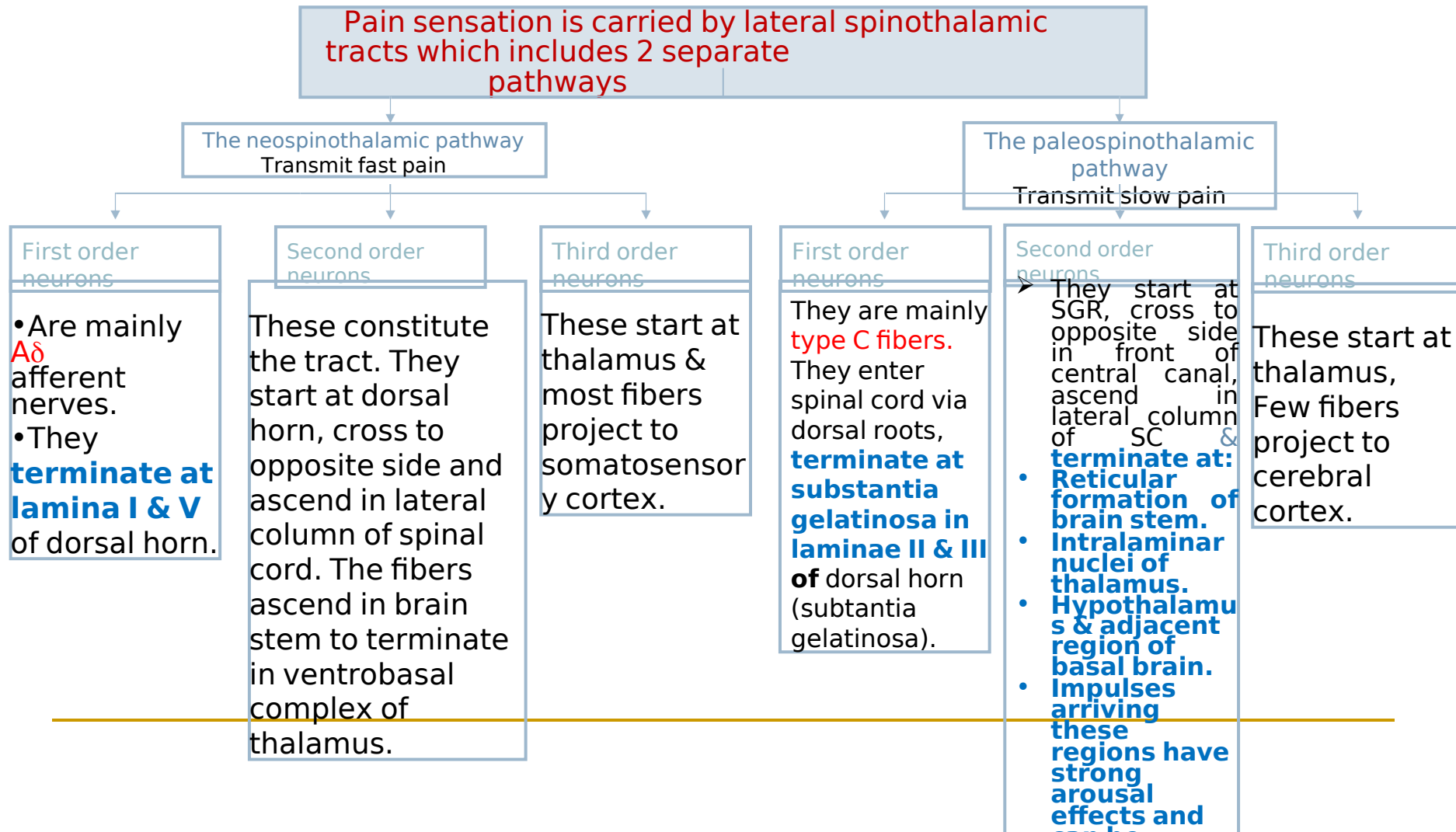




Nociceptive vs non-nociceptive pathways

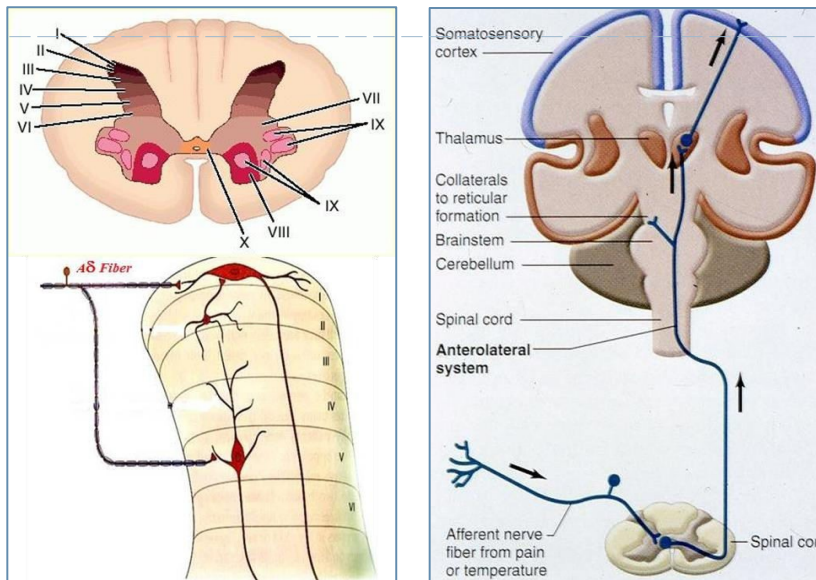


Pathway of pain

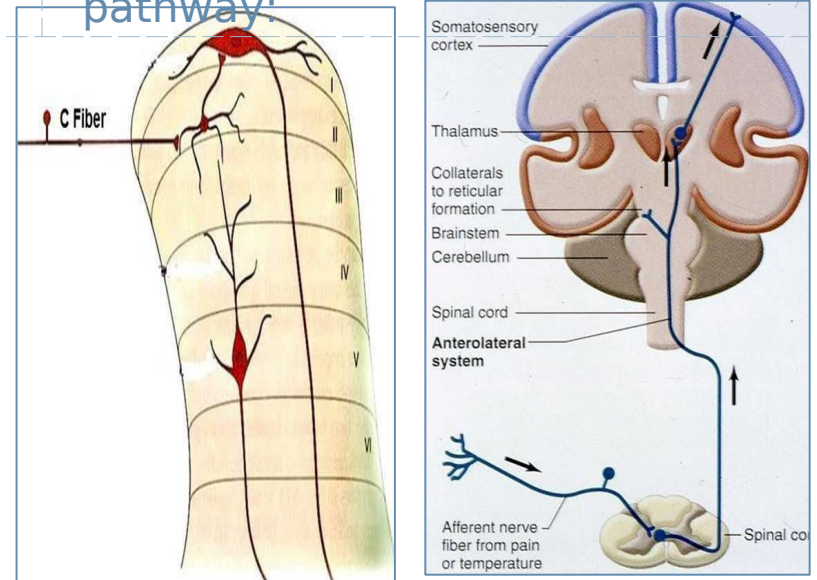


Most of the slow pain fibers project to reticular formation & then proceed to thalamus (posterior nuclei). Reticular system project to all parts of brain but specially to cerebral cortex therefore they cause arousal from sleep.

▶ The neospinothalamic pathway:



▶ The paleospinothalamic pathway:

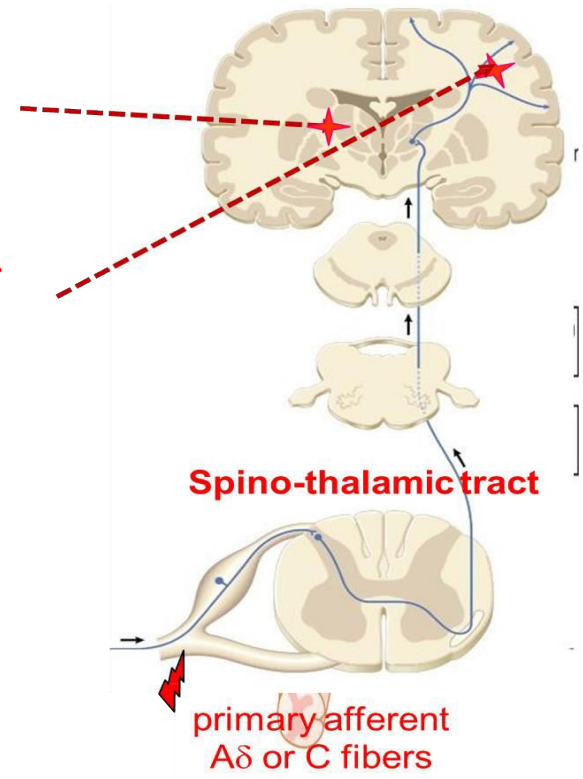


Role of thalamus and cerebral cortex in pain perception

Full perception of pain occurs when signals enter RF of brain stem, thalamus and basal regions.

Somatosensory cortex plays important role in topognosis i.e. **localization & interpretation of pain quality**.

Fast pain is localized better than slow pain because signals carried in neospinothalamic tract reach somatosensory cortex, while a small propotion of paleospinothalamic pathway reach there.



Pain Modulation

Modulation of pain occurs
• ***peripherally** at the nociceptor, in the spinal cord,*
or
• ***centrally** in supraspinal structures.*

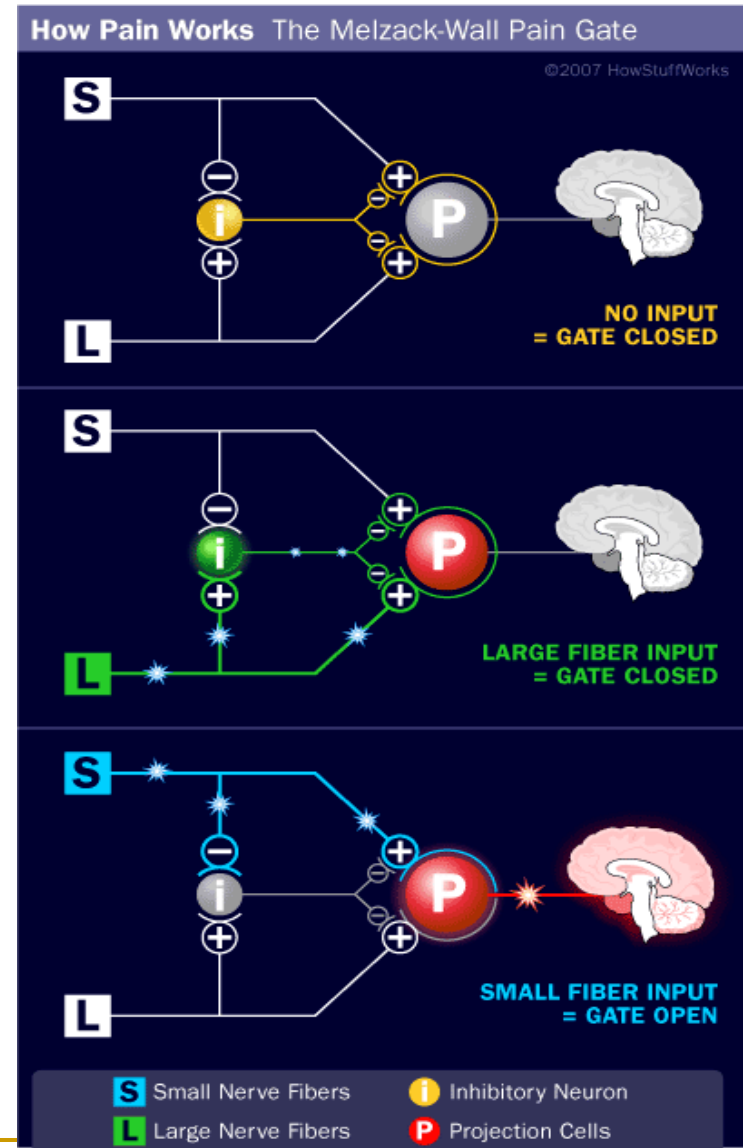
• *This modulation can either inhibit or facilitate
pain.*

Modulation at Spinal Level

Gate control theory

- *The Gate Control Theory devised by Patrick Wall and Ronald Melzack in 1965.*
 - *Pain is a function of the balance between the information traveling into the spinal cord through large nerve fibers and information traveling into the spinal cord through small nerve fibers.*
-

- *Without any stimulation, both large and small nerve fibers are quiet and the inhibitory interneuron (I) at SG blocks the signal in the projection neuron (P) at T cell that connects to the brain.*
- *The "gate is closed" and therefore NO PAIN.*



•With non-painful stimulation, large nerve fibers (like $A\beta$) are activated primarily. This activates the projection neuron (P),

BUT it ALSO activates the inhibitory interneuron (I) which then BLOCKS the signal in the projection neuron (P) that connects to the brain. The "gate is closed" and therefore

•NO PAIN.

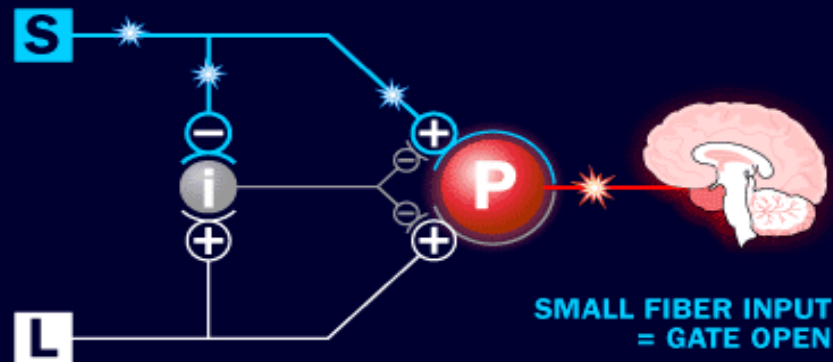
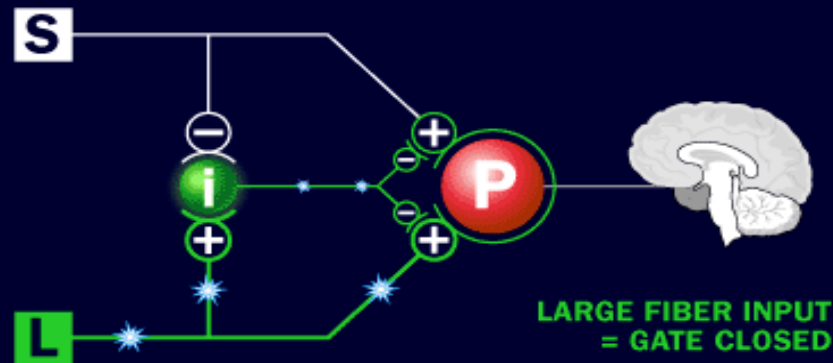
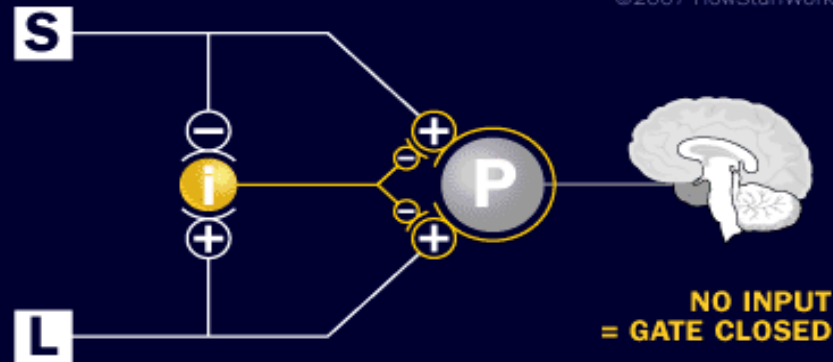
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- *With painful stimulation, small nerve fibers become active.*

They activate the projection neurons (P) and BLOCK the inhibitory interneuron (I).

Because activity of the inhibitory interneuron is blocked, it CANNOT block the output of the projection neuron that connects with the brain. The "gate is open", therefore, PAIN!!

How Pain Works The Melzack-Wall Pain Gate

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S Small Nerve Fibers

L Large Nerve Fibers

i Inhibitory Neuron

P Projection Cells

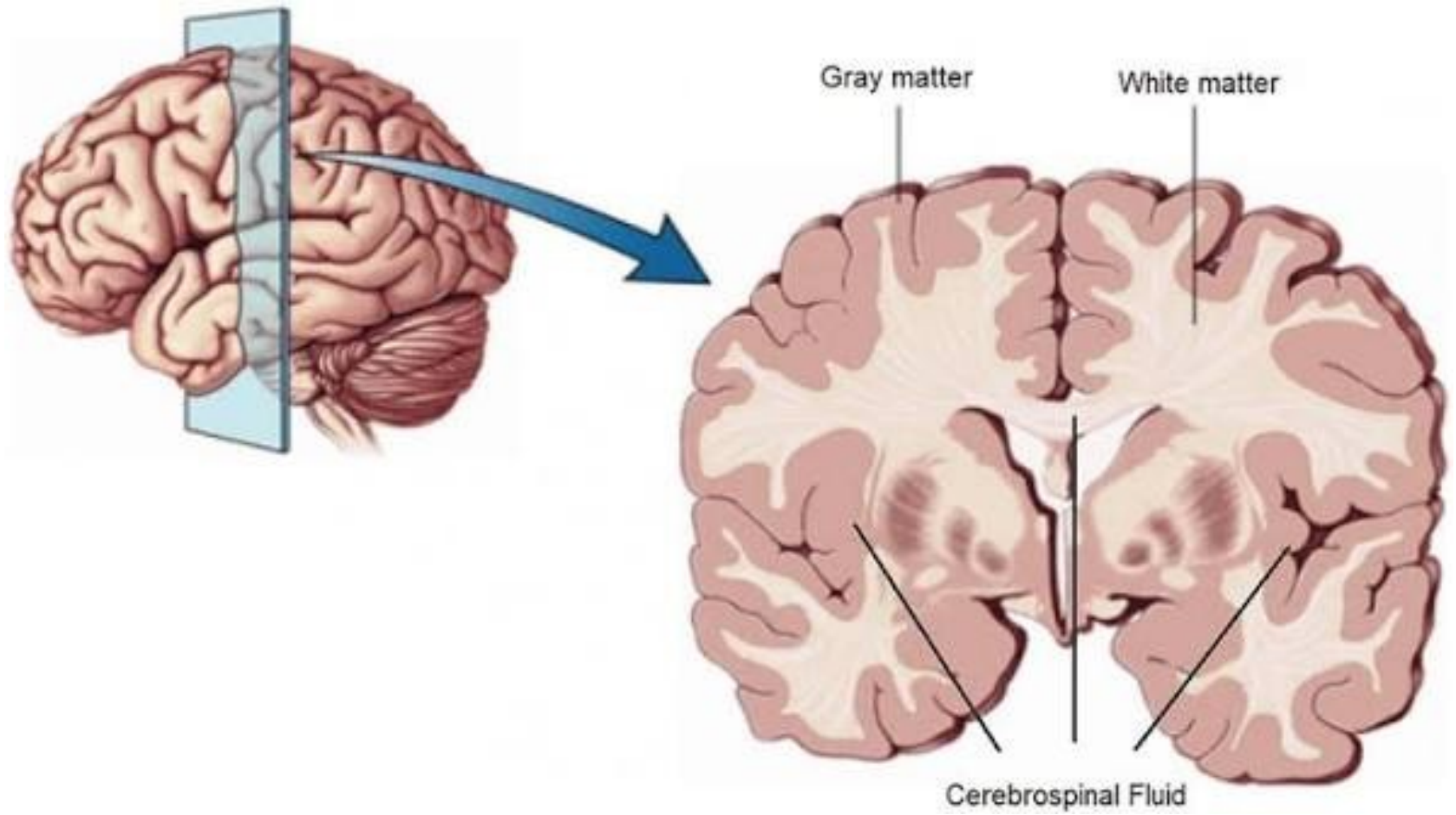
Gate control theory

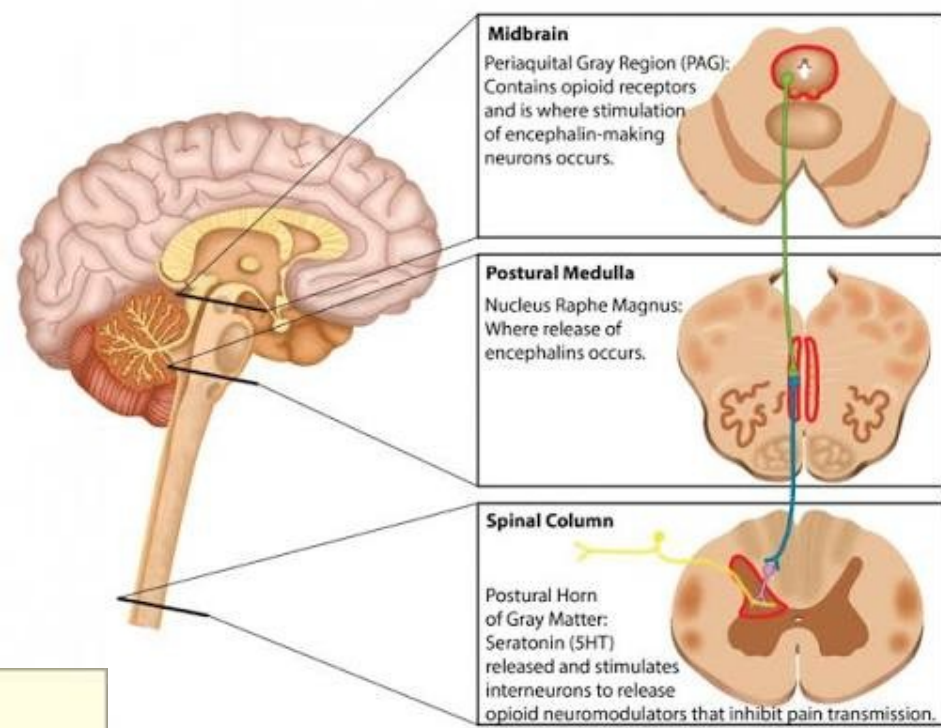
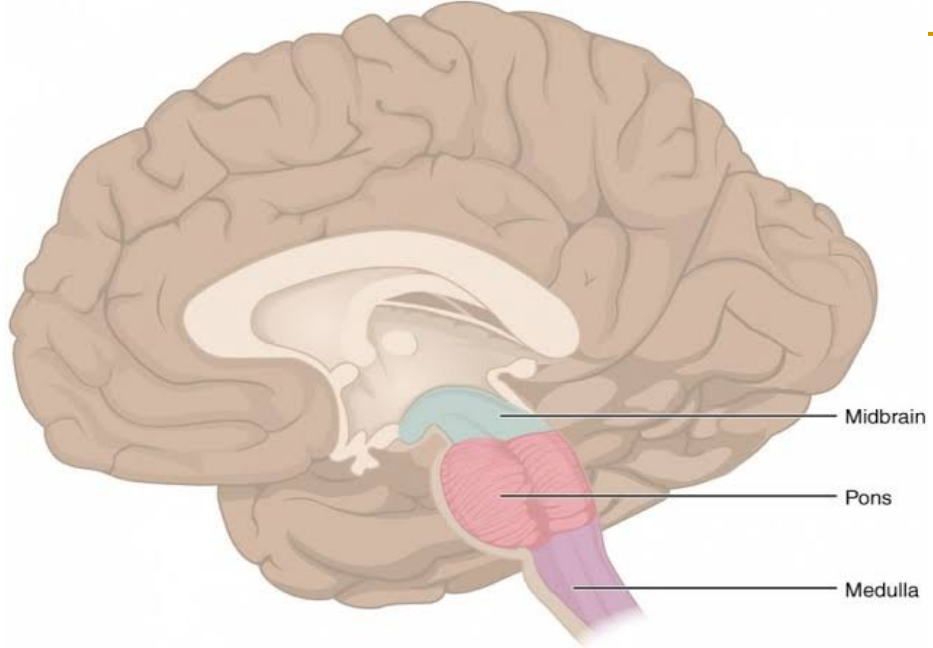
- *interaction between myelinated and nonmyelinated neurons occurs at inhibitory interneurons in **substantia gelatinosa** and at dorsal horn.*
- *The large myelinated afferents ($A\beta$ fiber) said to excite inhibitory interneurons and inhibit pain.*
- *The small fibers (type C and $A\delta$) inhibit the inhibitory interneurons.*

Descending Pain Modulation (endogenous opiate theory)

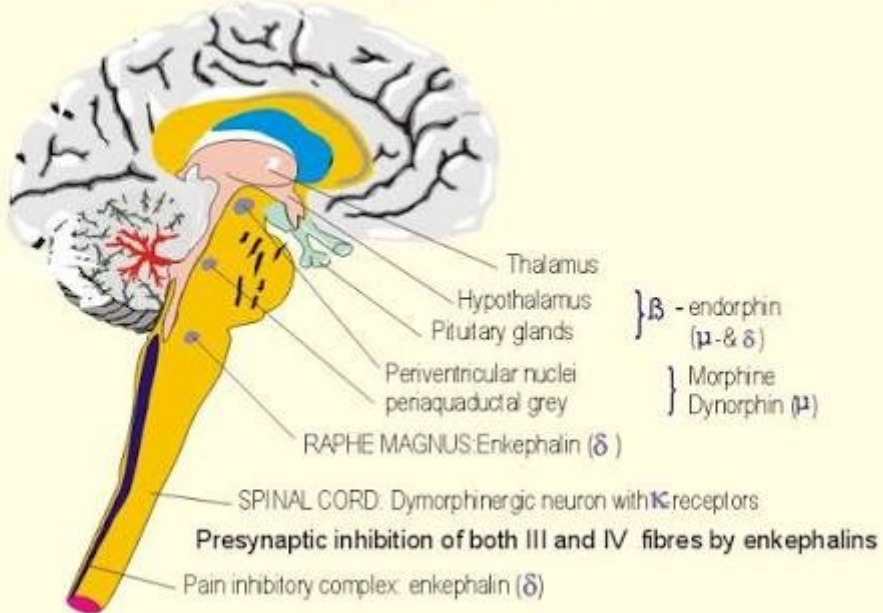
•The inhibitory effects from the higher centres come principally from periaqueductal grey matter (PAG) which is located at midbrain ,and raphe nucleus (RN) (located in the medulla).

These both have excitatory effects on the inhibitory interneurons of the substantia gelatinosa , and so have the ability to reduce the pain transmission.





Opiate Receptors In The CNS



Brainstem

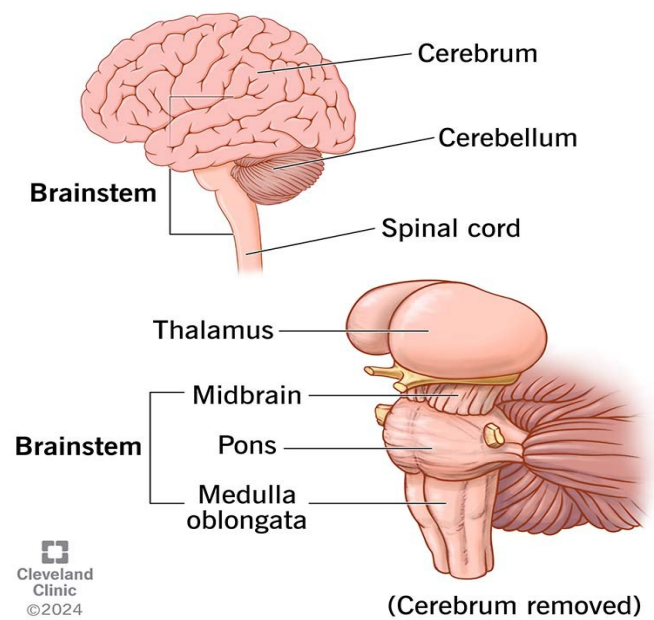
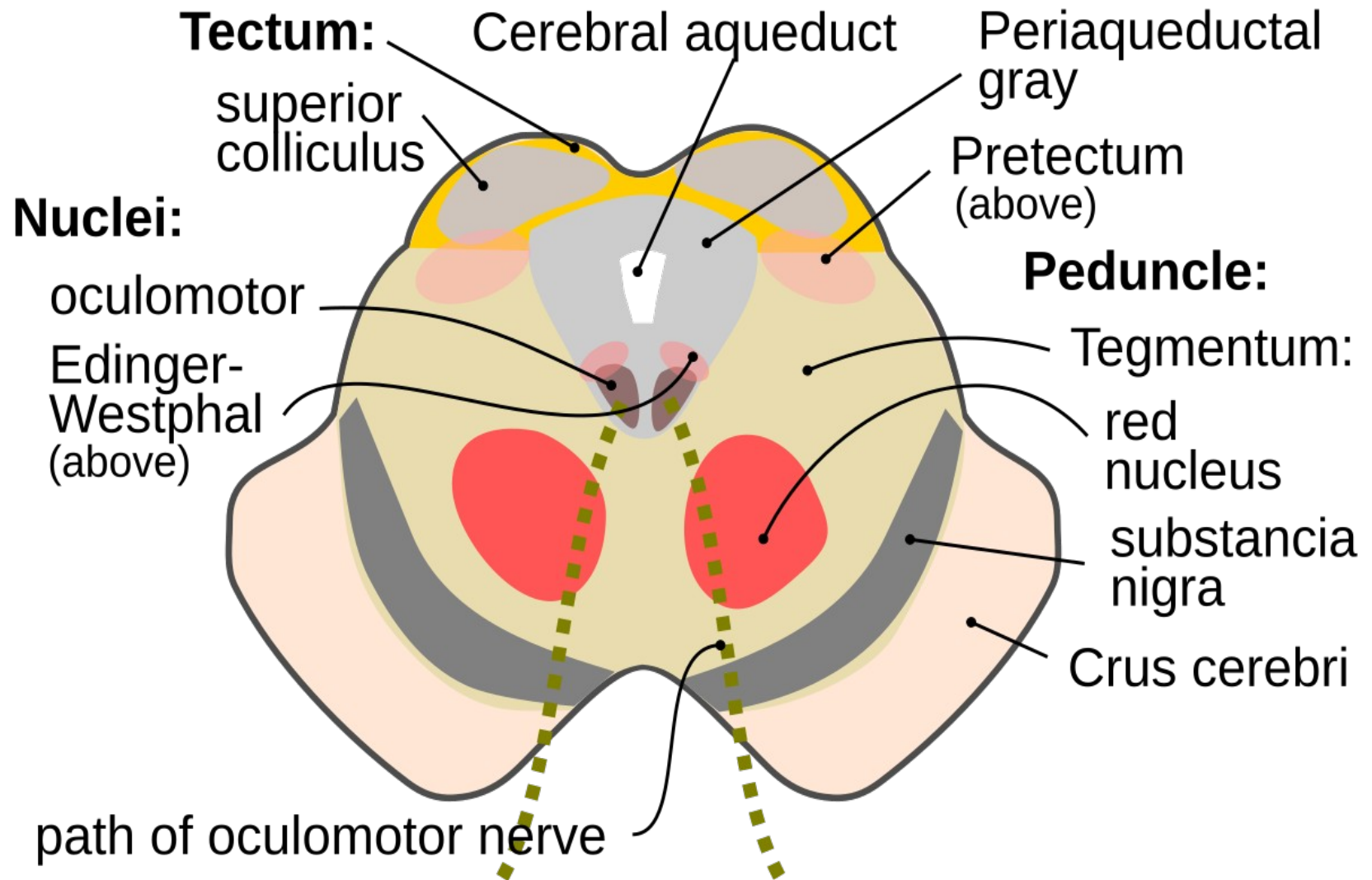


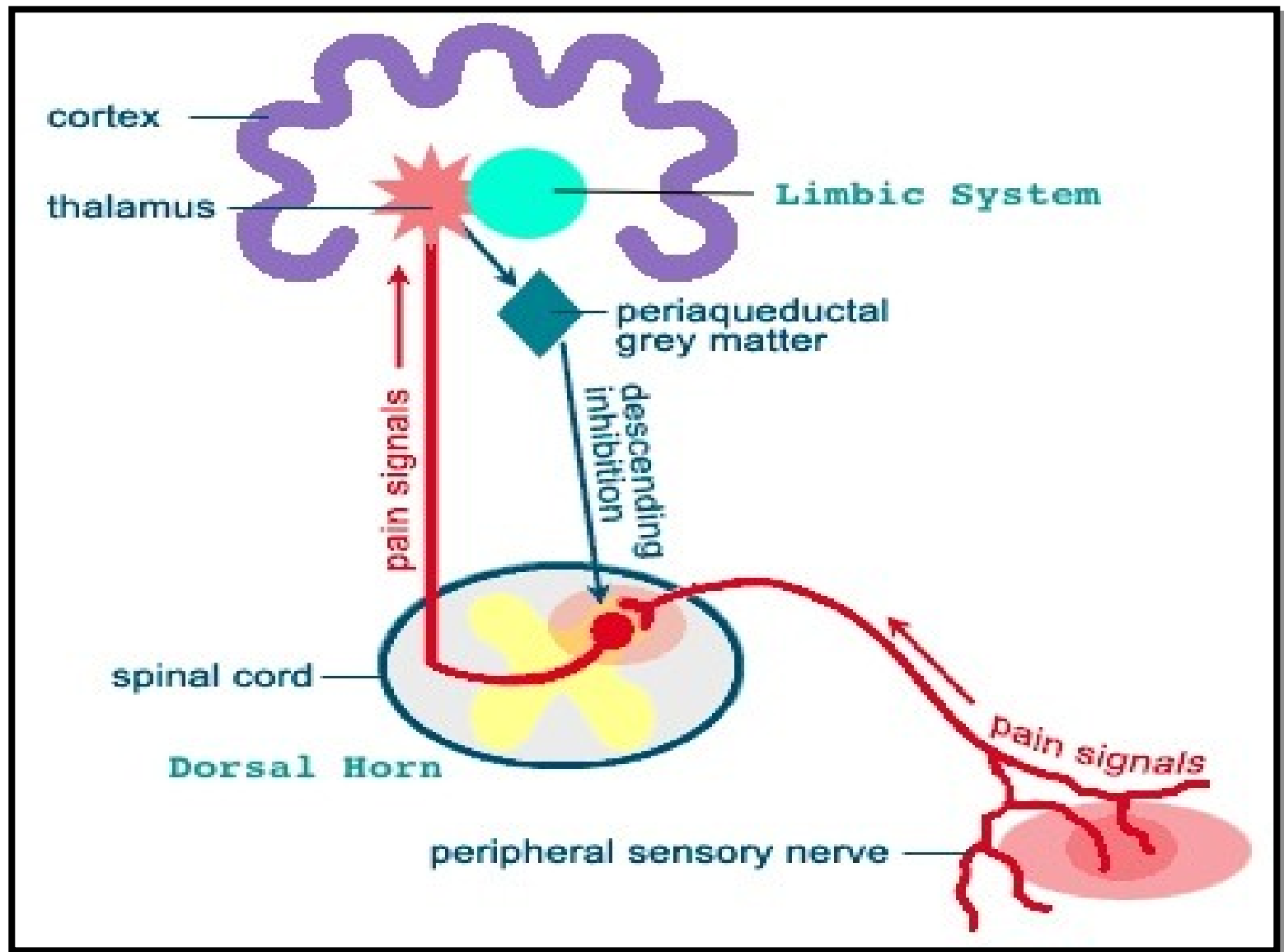
Fig. 3-10

KME



Endogenous Analgesics

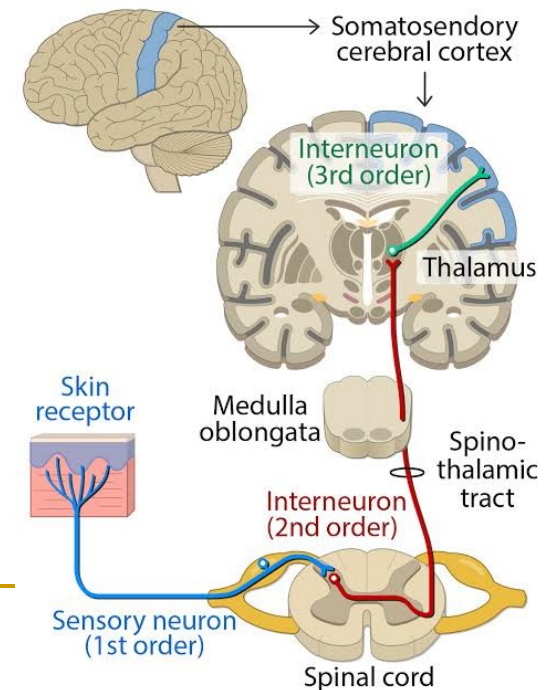
- *Opioid neuropeptides:*
enkephalins, endorphins, dynorphins, Endomorphins
- *The opioid peptides modulate nociceptive input in two ways:*
 - ***1) block neurotransmitter release by inhibiting Ca^{2+} influx into the presynaptic terminal,***
 - ***2) open potassium channels, which hyperpolarizes neurons and inhibits spike activity.***



Perception In the Thalamus and Cortex

Perception of pain varies depending on circumstances and past experiences.

Thalamocortical projections carry information on location, intensity and nature of pain.



PAIN PATHWAYS: FROM RECEPTORS TO BRAIN PERCEPTION

WHAT ARE PAIN PATHWAYS?

- Neural routes that transmit harmful stimuli
- Protect the body from injury
- Involve both peripheral and central nervous systems

SIGNAL PROCESSING

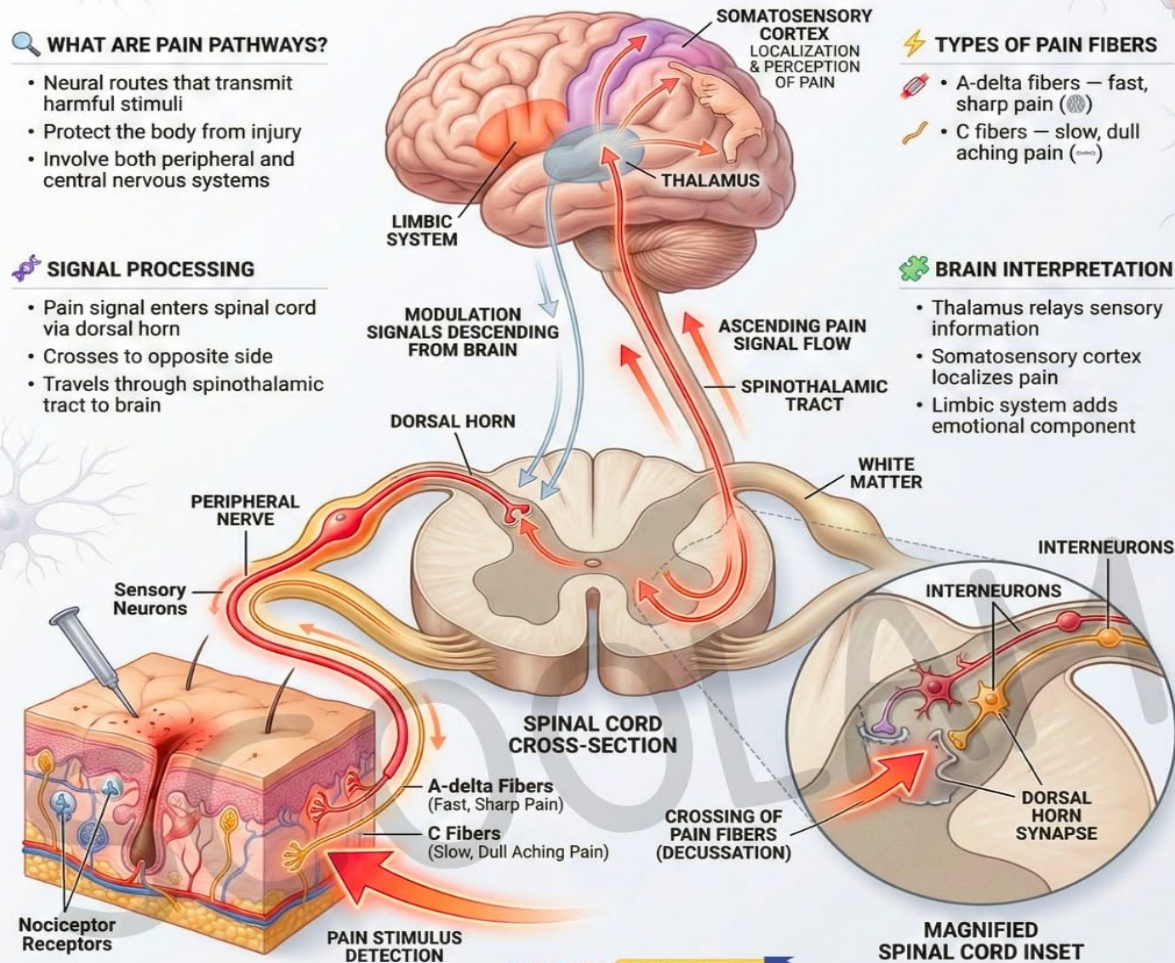
- Pain signal enters spinal cord via dorsal horn
- Crosses to opposite side
- Travels through spinothalamic tract to brain

TYPES OF PAIN FIBERS

- A-delta fibers — fast, sharp pain (⚡)
- C fibers — slow, dull aching pain (🐢)

BRAIN INTERPRETATION

- Thalamus relays sensory information
- Somatosensory cortex localizes pain
- Limbic system adds emotional component



PAIN TRANSMISSION SUMMARY (Table)

Step	Location
Detection	Nociceptor
Entry	Spinal cord
Relay	Thalamus
Perception	Cortex

CLINICAL RELEVANCE

- Chronic pain syndromes
- Neuropathic pain
- Analgesic drug targets
- Spinal cord injuries

BIOLOGICAL IMPORTANCE

- Warns against tissue damage
- Initiates protective reflexes
- Guides healing behaviors
- Essential survival mechanism

Qualities of pain

- ▶ (Phenomenon of double-pain).
- ▶ Fast / Sharp / immediate (1st) pain vs slow / diffuse / delayed (2nd) pain.

Fast pain

- Sharp, intense, pricking.
- It is felt within **0.1** sec. after stimulation.
Ex: pricking, cut with knife.
- Associated with reflex withdrawal.
- Usually somatic not visceral.
- Well localized and is mediated by A δ -fiber nociceptors in the peripheral nerves & centrally by **Neospinothalamic** Tract.
- **Occurs FIRST** upon stimulation of Mechanical and Thermal nociceptors.

Slow pain (or second)

- Burning, aching, throbbing “unbearable” diffuse, dull, or chronic pain.
- Felt after **1** sec or more.
- Associated with destruction of tissue.
- Can occur in skin or any internal organ / tissue.
- Poorly localized and is mediated by C-fiber nociceptors: misery (responsible for emotional aspect of pain).
- The noxious stimuli activates **10-20%** of the **A-delta** fibers and **50-80%** of the **C- fibers**.
- **Occurs SECOND** upon stimulation of Polymodal receptors.
- Chronic type of pain, transmitted by C fibers peripherally & centrally by **paleospinothalamic** Tract.

Characteristics of nerve fibers

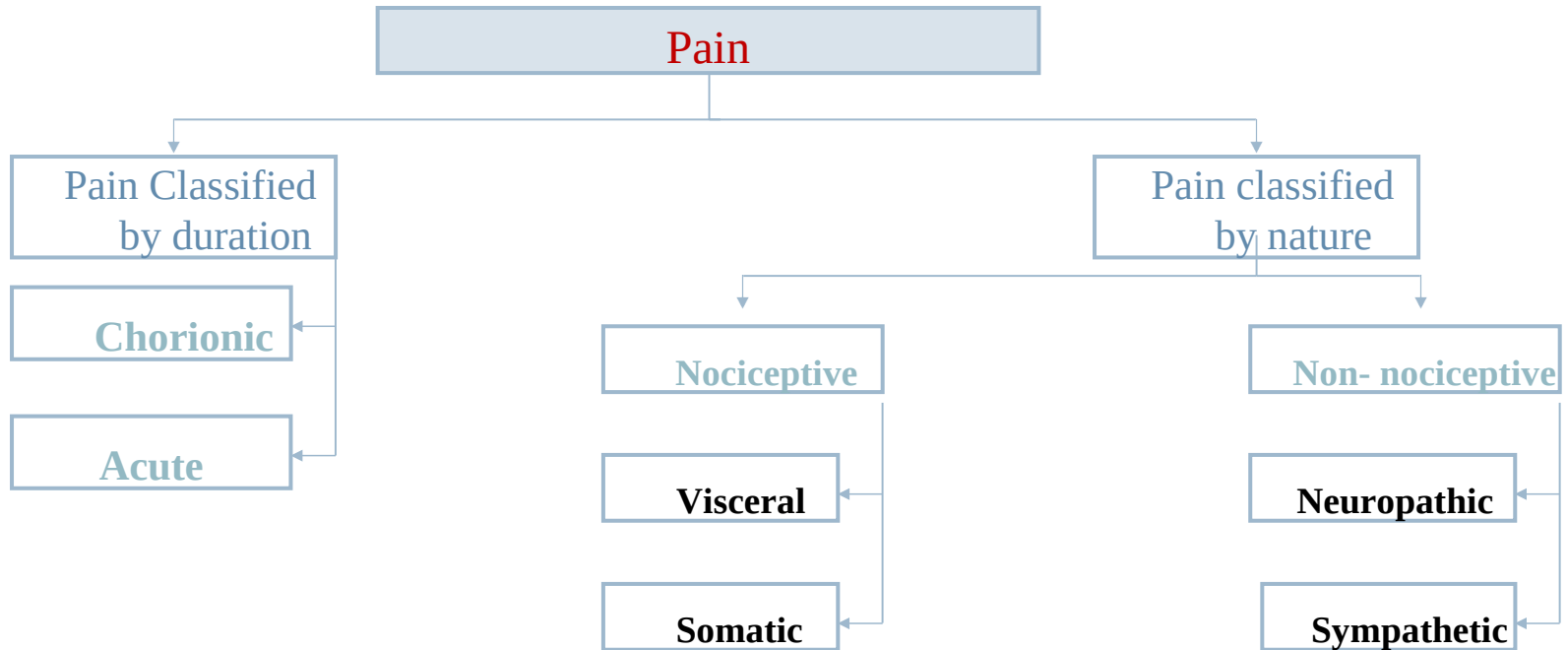
- **Characteristics of A δ fiber:**

- **Myelinated.**
- **Diameter fine 2 - 5 μm .**
- **12 - 30 m/sec. conduction velocity.**
- **Terminate at I and V laminae.**
- **Neurotransmitter – is glutamate.**

- **Characteristics of C fibers:**

- **Non-Myelinated.**
- **Diameter fine 0.4 – 1.2 μm .**
- **0.5 – 2 m/sec. conduction velocity.**
- **Terminate at II and III laminae,**
- **Neurotransmitter – is Substance- P.**

Classification of pain



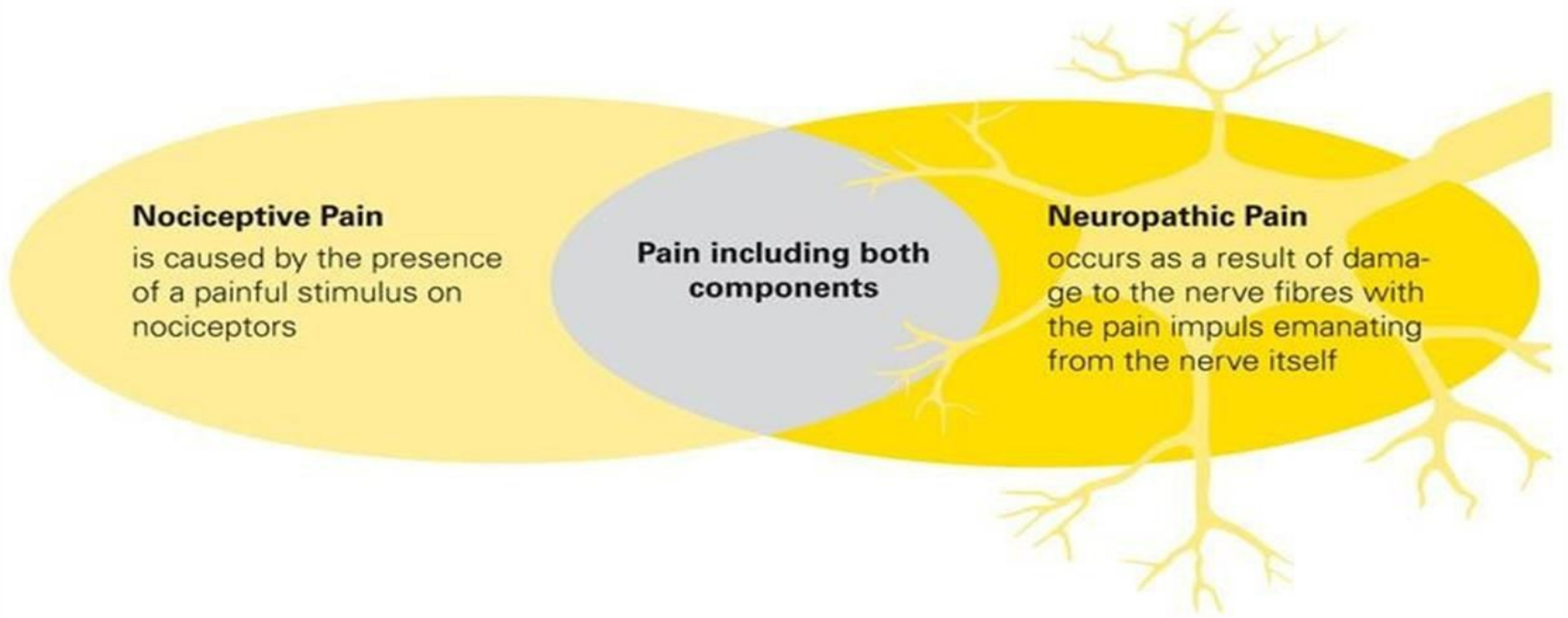
Nociceptive Pain

is caused by the presence of a painful stimulus on nociceptors

Pain including both components

Neuropathic Pain

occurs as a result of damage to the nerve fibres with the pain impulses emanating from the nerve itself



Chronic and Central Pain

- **Chronic Pain**: *if healing never occurs or as a result of prolonged or intense pain, persistent even after tissue repair*

- **Central Pain** *(various forms)*

Damage in CNS (spinothalamic tract, thalamus, somatosensory cortex, lesion of inhibitory Pathways)

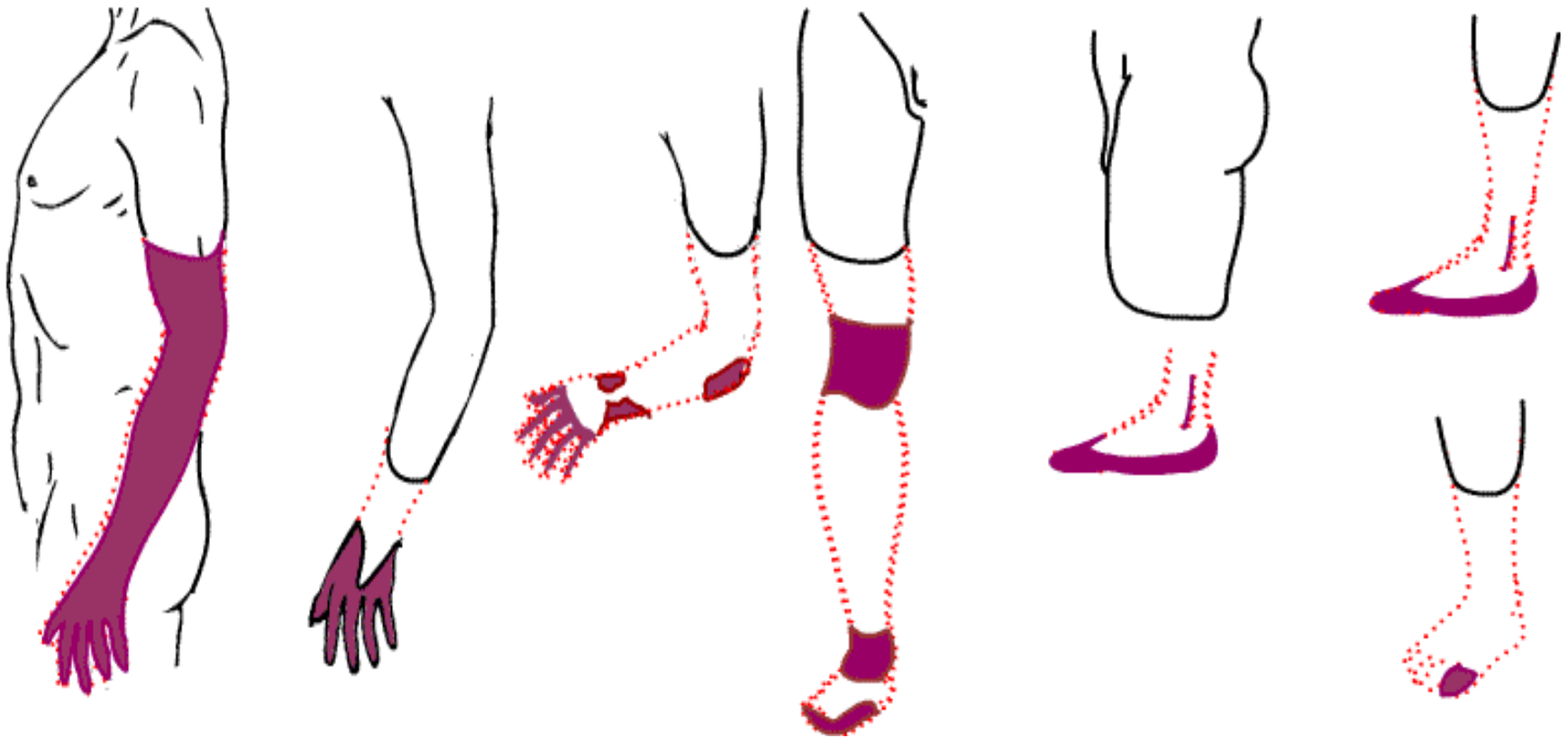
Often insensitive to opioids but anti-epileptic drugs can alleviate

i.e. central pain is a specific type of neuropathic pain

Some Specific Manifestations (types) of Pain

- Neuropathic pain is caused by damage or disease affecting any part of the nervous system involved in bodily feelings (the somatosensory system). Peripheral neuropathic pain is often described as "burning", "tingling", "electrical", "stabbing", or "pins and needles" .
- Phantom pain is pain felt in a part of the body that has been lost or from which the brain no longer receives signals. It is a type of neuropathic pain. it is a common experience of amputees.

:Neuropathic pain



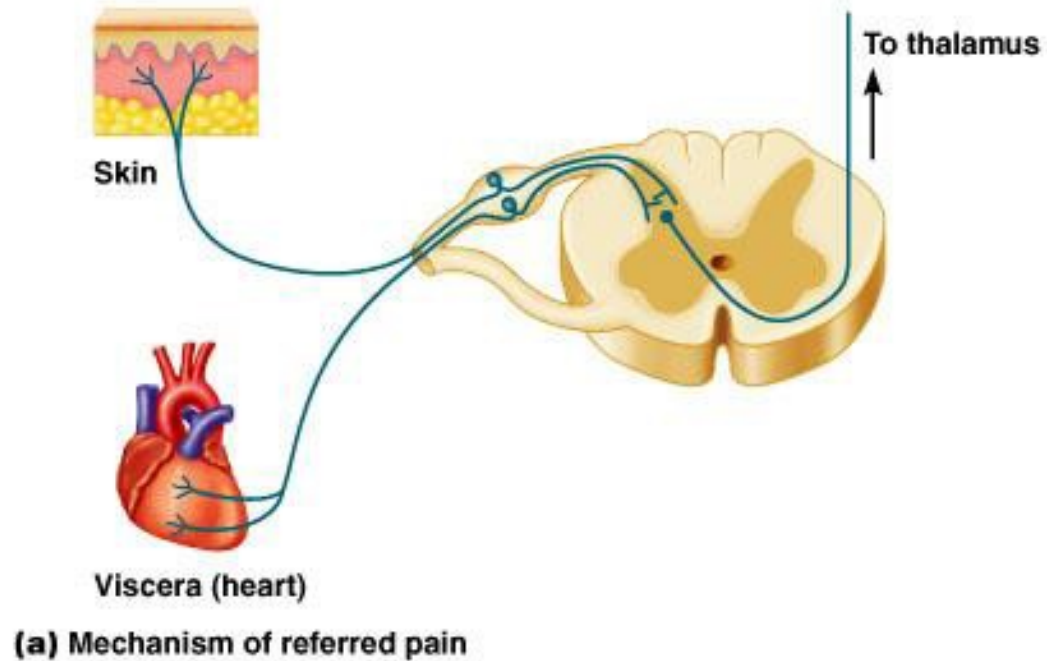
Phantom Limb Phenomenon

Suggested mechanisms:

- Peripheral: damage to nociceptive fibres and abnormal discharges from the remaining cut ends of nerves which grow into nodules called neuromas
- Spinal: sprouting of nociceptive fiber terminals to make **new connections, second order neurone sensitization and overactive spinal neurons.**
- Higher Centres: abnormal flow of signals through the somatosensory cortex, or from **burst-firing neurons** in the thalamus
- **Unresponsive to morphine**

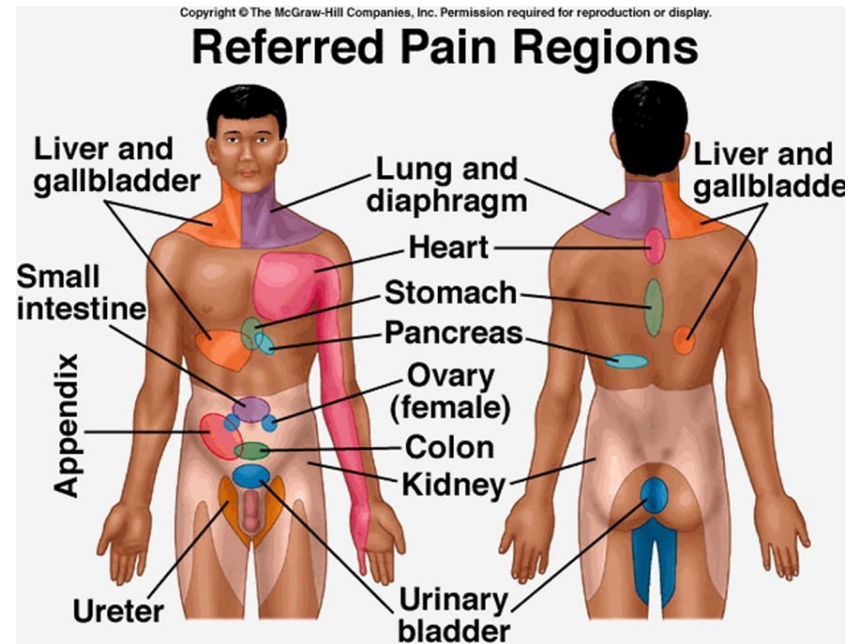
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- Migraine: *Migraine is also a central type of pain, it is not sensitive to morphine, in which Abnormality of cerebral vascular function Vasospasm followed by smooth muscle relaxation (vasodilatation)*
 - Referred pain: *Referred pain arises because of the convergence of nociceptive and cutaneous fibres in the dorsal horn of the spinal cord, so that pain arising in visceral structures may produce sensations associated with areas of the body surface.*
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Referred Pain



Referred Pain

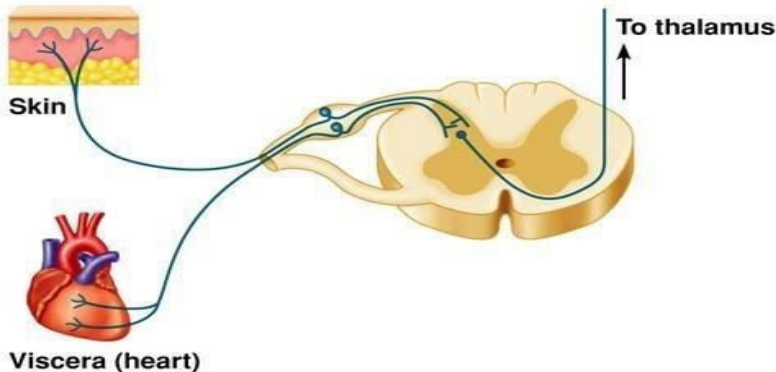
- Visceral pain that is felt at a somatic structure that can be far away from the origin site.
- Poorly localized and is not identical in all people.
- In some people heart pain can be referred to right (not left) arm or neck.
- It is most frequent with visceral pain & deep somatic pain **but cutaneous pain is not referred.**
- Pain is referred according to **dermatomal rule.**



Mechanism of referred pain

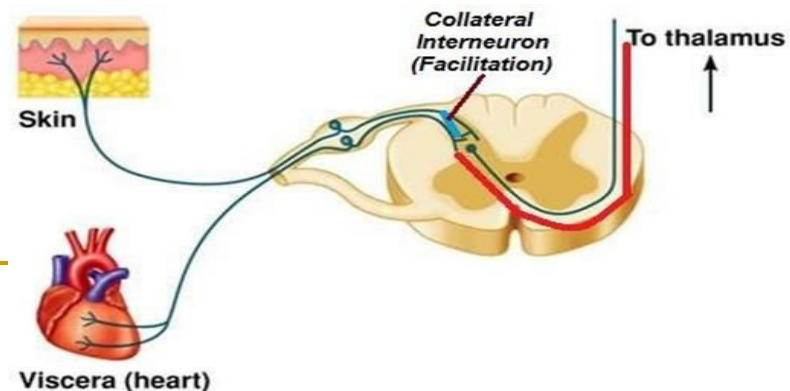
1. Convergence theory:

- Afferent pain fibers from skin area & diseased viscera that develop from same embryonic segment converge on same 2nd order neuron and finally stimulate the same cortical neuron.
- The brain interprets the information coming from visceral nociceptors as having arisen from cutaneous nociceptors, because this is where nociceptive stimuli originate more frequently



2. Facilitation theory:

- Pain fibers from skin are always carrying impulses, not enough to produce pain.
- Impulses from diseased viscus pass through afferents which give collaterals to ST neurons receiving pain fibers from skin.
- As a result, ST neurons' excitability is raised (they are facilitated) to reach a threshold level.
- The signals reaching the brain are projected to skin area and pain is felt in skin dermatome.



Examples of referred pain

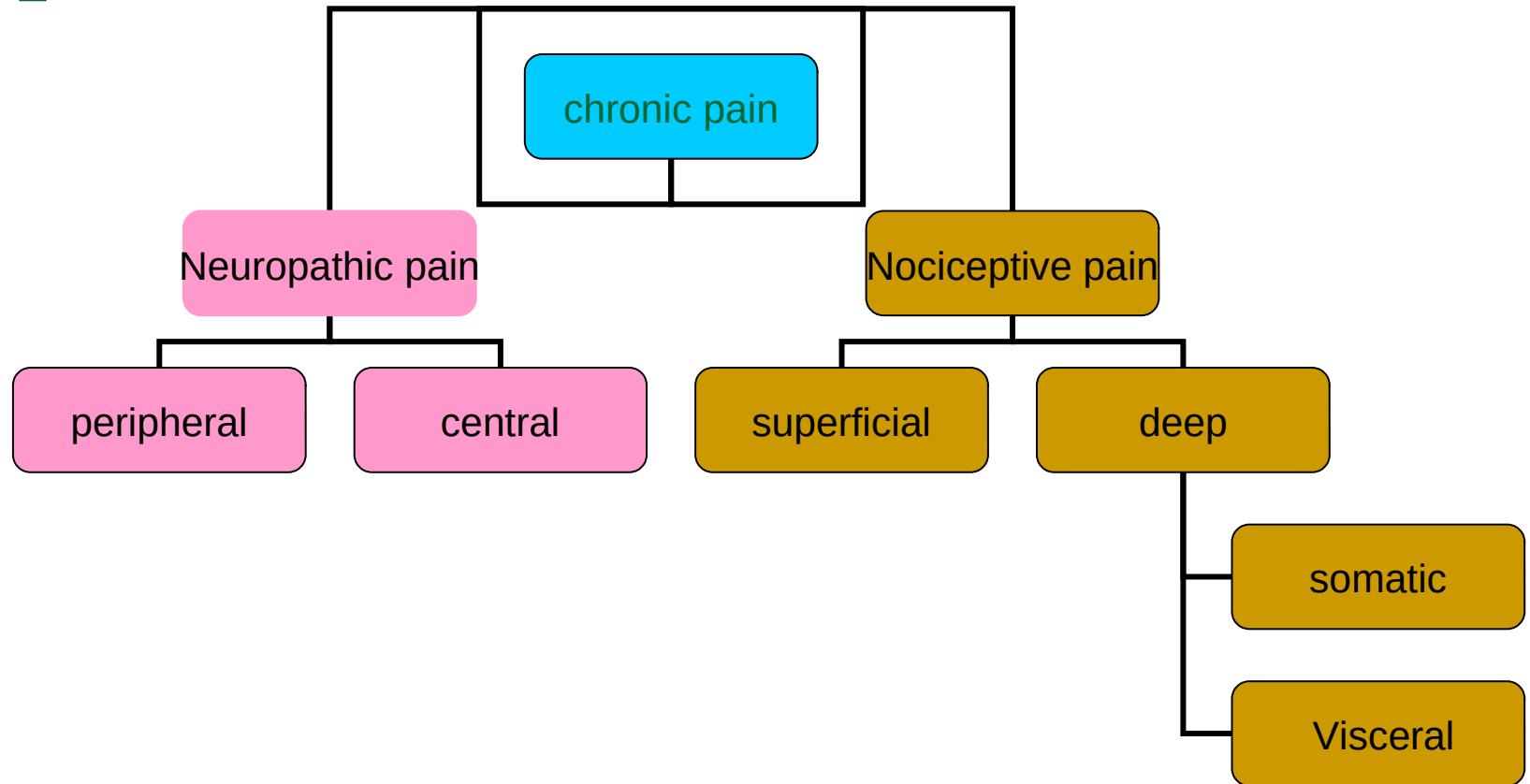
- Cardiac pain is referred to the jaw, left shoulder & inner side of left arm.
 - Pain of appendicitis is referred to periumbilical region.
 - Pain from the ureter is referred to testicular region.
-

Chronic Pain

?What is Chronic Pain

- Chronic pain is persistent or recurrent pain, lasting beyond the usual course of acute illness or injury, or more than 3 - 6 months, and adversely affecting the patient's well-being.
 - **Pain that continues when it should not.**
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Classification of chronic pain



Nociceptive vs Neuropathic Pain

Nociceptive Pain

Caused by activity in neural pathways in response to potentially tissue-damaging stimuli

Postoperative pain

Mechanical low back pain

Sports/exercise injuries

Sickle cell crisis

Arthritis

Neuropathic Pain

Caused by primary lesion or dysfunction in the

Postherpetic neuralgia

Neuropathic low back pain

Distal polyneuropathy (eg, diabetic, HIV)

Trigeminal neuralgia

Pathophysiology of chronic pain

- Persistent activation nociceptive transmission to the dorsal horn may induce a ***pain wind-up phenomenon***.

(Pain wind-up: is a frequency-dependent increase in the excitability of spinal cord neurons, evoked by electrical stimulation of afferent C-fibers. It occurs due to sustained state of excitation of the projection neurons induced by glutamate activation of NMDA receptors).

Pathophysiology of chronic pain

- This induces pathological changes that lower the threshold for pain signals to be transmitted.
 - It may generate non-nociceptive nerve fibers to respond to pain signals.
 - In chronic pain this process is difficult to reverse or eradicate once established.
-

Central pain

- **Central pain state:** a neurological condition caused by *damage or malfunction in the NS* which causes a sensitization of the pain system.
- It is almost always generated by **wind-up of the projection neurons of the spinothalamic and spinoreticular pathways**. The extent of pain and the areas affected are related to the cause of the injury.

Central (thalamic) pain syndrome

Dejerine–Roussy syndrome

- is a condition developed after a thalamic stroke, a stroke causing damage to the thalamus.
- An initial lack of sensation and tingling in the opposite side of the body.
- Weeks to months later, numbness can develop into severe and chronic pain that is not proportional to an environmental stimulus, called **dysaesthesia or allodynia**.

Chronic Pain Syndrome

- **Chronic Pain Syndrome** (CPS) is a common problem that is complex in nature of poor etiology and poor response to therapy.
 - Most consider ongoing pain of 3~6 months are diagnostic.
 - Two or more co-existing pain conditions or widespread generalized pain.
 - Although CPS has been known to resolve completely with (and very rarely without) treatment, it is uncommon for full relief to be achieved
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